

MulFS-CAP: Multimodal Fusion-Supervised Cross-Modality Alignment Perception for Unregistered Infrared-Visible Image Fusion

Huafeng Li [✉], *Member, IEEE*, Zengyi Yang [✉], Yafei Zhang [✉], Wei Jia [✉], *Member, IEEE*, Zhengtao Yu [✉], and Yu Liu [✉], *Senior Member, IEEE*

Abstract—In this study, we propose Multimodal Fusion-supervised Cross-modality Alignment Perception (MulFS-CAP), a novel framework for single-stage fusion of unregistered infrared-visible images. Traditional two-stage methods depend on explicit registration algorithms to align source images spatially, often adding complexity. In contrast, MulFS-CAP seamlessly blends implicit registration with fusion, simplifying the process and enhancing suitability for practical applications. MulFS-CAP utilizes a shared shallow feature encoder to merge unregistered infrared-visible images in a single stage. To address the specific requirements of feature-level alignment and fusion, we develop a consistent feature learning approach via a learnable modality dictionary. This dictionary provides complementary information for unimodal features, thereby maintaining consistency between individual and fused multimodal features. As a result, MulFS-CAP effectively reduces the impact of modality variance on cross-modality feature alignment, allowing for simultaneous registration and fusion. Additionally, in MulFS-CAP, we advance a novel cross-modality alignment approach, creating a correlation matrix to detail pixel relationships between source images. This matrix aids in aligning features across infrared and visible images, further refining the fusion process. The above designs make MulFS-CAP more lightweight, effective and explicit registration-free. Experimental results from different datasets demonstrate the effectiveness of our proposed method and its superiority over the state-of-the-art two-stage methods.

Index Terms—Cross-modality alignment perception, unregistered infrared-visible image fusion, learnable modality dictionary, registration and fusion.

Received 3 February 2024; revised 15 October 2024; accepted 24 January 2025. Date of publication 28 January 2025; date of current version 3 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62161015, Grant 62176081, Grant U23A20294, Grant 62076086, and Grant 62476077 and in part by Yunnan Fundamental Research Projects under Grant 202301AV070004. Recommended for acceptance by M. Cho. (*Corresponding author: Yu Liu.*)

Huafeng Li, Zengyi Yang, Yafei Zhang, and Zhengtao Yu are with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China (e-mail: lhfchina99@kust.edu.cn; zengyiyang0211@gmail.com; zyfeimail@163.com; ztyu@hotmail.com).

Wei Jia is with the School of Computer Science and Information Engineering, Hefei University of Technology, Hefei 230009, China (e-mail: jiawei@hfut.edu.cn).

Yu Liu is with the Department of Biomedical Engineering, Hefei University of Technology, Hefei 230009, China (e-mail: yuliu@hfut.edu.cn).

The source code of our method is available at <https://github.com/YR0211/MulFS-CAP>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TPAMI.2025.3535617>, provided by the authors.

Digital Object Identifier 10.1109/TPAMI.2025.3535617

I. INTRODUCTION

DUE to the differing mechanisms of infrared and visible sensors in imaging, images captured from the same scene using these sensors often contain a rich set of complementary information. By integrating such information into a single image, the comprehensiveness of the scene's depiction can be significantly enhanced. The resulting fused image can then be utilized in various subsequent computer vision tasks. Consequently, the fusion of infrared and visible images has found extensive applications in diverse fields such as military surveillance and reconnaissance, intelligent driving, environmental monitoring, and aerospace.

In recent years, there have been extensive researches on infrared-visible (IR-VIS) image fusion. These studies cover various aspects, including the fusion of IR-VIS images at different resolutions [1], [2], multisource image fusion with a unified framework [3], [4], [5], [6], [7], [8] and IR-VIS image fusion combined with high-level vision tasks [9], [10], [11], [12], [13], [14], [15], etc. Although these methods have made significant progress in their respective areas, they all assume that the images to be fused are strictly registered. Nevertheless, images obtained in real-world scenarios often do not satisfy this condition, making such algorithms incapable of fusing unregistered source images directly in practical applications.

To address this issue, unregistered IR-VIS image fusion has attracted attention in the academic community [16], [17], [18], [19], [20], [21], [22], [23], [24], leading to the proposal of effective solutions. To address the issue of unregistered IR-VIS image fusion, most existing methods follow a two-stage processing paradigm. The first stage performs a registration algorithm on the input source images to achieve pixel-level alignment. Subsequently, the results of the registration are fused in the second stage using an image fusion method, as shown in Fig. 1(a). Particularly, ReCoNet [16] and UMF-CMGR [17] adopt two independent stages of registration and fusion to achieve the fusion of unregistered images. Despite their effectiveness, these approaches suffer from a crucial drawback: the independent training of models for each task prevents synergistic enhancement, often leading to suboptimal fusion results. To address these limitations, RFNet [18] and MURF [19] utilize a two-stage method that incorporates joint training. This method facilitates mutual enhancement of the fusion and registration processes by taking into account the quality of fusion results during the

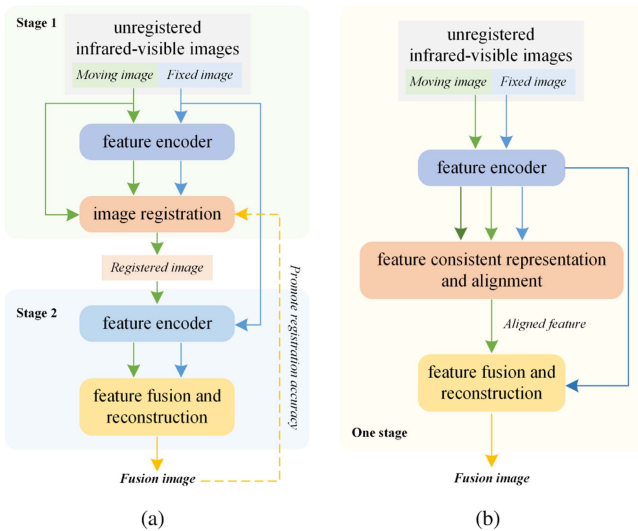


Fig. 1. The difference of the proposed method and the existing methods. (a) The existing two-stage fusion methods require a complete and explicit registration, whereas (b) the proposed method operates without the need for complete and explicit registration. The dashed line in (a) represents the optimization of the registration model based on feedback from the fusion results.

training of the registration model. Such an approach not only enhances the accuracy of registration but also indirectly improves the overall quality of the fusion outcomes. Nonetheless, these methods still cannot escape the shortcomings inherent in the two-stage processing paradigm. The image registration employed in the two-stage methods constitutes a full and explicit registration process, capable of operating independently from the fusion model. However, this independence significantly increases the overall complexity of the fusion system, particularly when dealing with unregistered source images, posing challenges for deployment on platforms with limited computational resources.

Furthermore, the significant differences in modality between infrared and visible images pose a major challenge for effective cross-modality registration. To address this issue, UMF-CMGR [17] and RFNet [18] introduce the approach of generating an image in one modality based on the input from another, using this generated image to facilitate registration. While this approach can mitigate the impact of modality discrepancy, the effectiveness of image registration heavily relies on the quality of the generated results. To overcome the challenges posed by modality discrepancy in cross-modality registration, a common strategy is to learn shared features from the images intended for fusion. Specifically, MURF [19] introduce a method to extract modality-shared features from IR-VIS images through contrastive learning. Similarly, SuperFusion [20] employ an encoder with shared parameters and consistency constraints for this purpose. However, precise registration often faces challenges due to the limited shared features across modalities.

After analyzing the limitations of current methods, we argue that integrating image registration into the fusion process could effectively overcome some inherent drawbacks of the two-stage method. This idea, if realized, would enable the completion of tasks that typically require a two-stage process within a unified single-stage fusion framework. However, a significant challenge arises from the conflicting requirements that registration and

fusion impose on the feature extraction. Image registration necessitates aligning corresponding pixels of unregistered source images in spatial positions, which often requires consistent feature extraction from the IR-VIS input images. Without this, registration becomes more challenging. By contrast, the goal of IR-VIS image fusion is to maximize the integration of complementary information from the source images. This requires the feature extraction network used for fusion tasks to capture as much complementary information as possible, enhancing the comprehensiveness of the scene description. Achieving these diverse requirements with a single encoder within the same framework presents a notable challenge. This incompatibility is an underlying reason why combining both tasks in a single-stage framework is difficult when dealing with the fusion of unregistered IR-VIS images.

In this study, we propose a novel and more effective solution, termed MulFS-CAP, to tackle the challenges previously mentioned. This design allows the processes, typically accomplished in two stages, to be achieved in a single stage by enabling both alignment and fusion to share a shallow feature encoder, as depicted in Fig. 1(b). More specifically, within this single-stage framework, we introduce a learnable modality dictionary specifically designed to enhance single-modality features with the information provided by the dictionary itself. This strategy effectively creates consistent features, meeting the requirements of cross-modality image registration. With the support of the learned modality dictionary, MulFS-CAP balances the two conflicting tasks of registration and fusion, integrating them seamlessly into a single-stage framework. Furthermore, by compensating features with the modality dictionary, we propose a cross-modality alignment perception mechanism. This mechanism generates a correlation matrix that represents the pixel correspondence of the source images, enabling spatial alignment of the features to be fused based on the matrix. Utilizing these aligned multimodal features allows us to reconstruct fusion results that are unaffected by misalignment issues. The aforementioned designs make the proposed method more lightweight, effective and explicit registration-free, as demonstrated in Fig. 1. The contributions and the advantages of the proposed method are summarized in the following points:

- *New Perspective:* Departing from traditional two-stage methods of unregistered image fusion, this study focuses on achieving the fusion of unregistered IR-VIS images within a single-stage framework. This approach is designed to reduce the extensive parameterization typically required by explicit registration in models. To the best of our knowledge, this represents the first attempt to jointly address two previously incompatible tasks within a single-stage framework. The implementation of this concept can effectively overcome the limitations associated with two-stage methods and bring new ideas to the design of solutions for the fusion of unregistered IR-VIS images.
- *Novel Approach:* By integrating a learnable modality dictionary, we address the conflicting feature requirements for registration and fusion, enabling seamless integration of implicit registration and fusion within a single-stage, explicit registration-free framework. Furthermore, we develop an efficient feature alignment perception approach,

which analyzes input feature differences to establish feature correlations, constructing a correlation matrix for cross-modality alignment. This approach not only eliminates the need for explicit image registration but also reduces the model's parameters.

- *State-of-the-art Results:* We assessed our proposed method across various IR-VIS datasets to validate its performance. According to the results of the experiments, our method surpasses existing state-of-the-art methods in terms of visual quality and objective evaluation metrics, while requiring fewer model parameters.

The rest of this paper is organized as follows: Section II provides a comprehensive review of related work. Section III elaborates on the proposed method in detail. Section IV presents the experimental results and effectiveness analysis. Finally, Section V concludes this paper.

II. RELATED WORK

Current for IR-VIS image fusion can broadly be divided into two categories: registered fusion and unregistered fusion. In contrast to the well-established field of image fusion research on registered images, the exploration of fusion techniques for unregistered images is still in its infancy.

A. Registered IR-VIS Image Fusion

The primary goal of these methods is to enhance fusion results by meticulously designing networks for feature extraction. These methods can be categorized based on the network type used in feature extraction, primarily including Convolutional Neural Network (CNN)-based methods [5], [25], [26], [27], [28], [29], CNN-Transformer-based methods [2], [30], [31], [32], [33], [34], [35], [36], [37] and Generative Adversarial Network (GAN)-based methods [2], [30], [31], [32], [33], [34], [35], [36], [37]. In most CNN-based methods for IR-VIS image fusion, a series of operations like convolution, activation, and pooling are performed to extract features from source images. These features are then fused and reconstructed to produce the final fusion outcome. Since the quality of the extracted features greatly impacts the reconstruction, the design of the feature extraction network is a focal point in most CNN-based fusion methods. Within the CNN framework, it is now common to introduce dense connections [25], skip connections [26], and nested connections [27] between convolutional layers, as well as embedding attention mechanisms [28], [29], to enhance the quality of the extracted features. Notable research employing these techniques includes U2Fusion [3], DenseFuse [25], IFCNN [5] and CoCoNet [38].

The limited receptive field of CNNs hinders their ability to capture features with distant relationships, thus constraining their expressive capabilities. By contrast, Transformers demonstrate a superior ability to model long-range dependencies, rendering them more effective for extracting global features in images. This attribute has notably increased the adoption of Transformers in the field of image fusion. For instance, PPTFusion [30] integrates the Transformer into image fusion, utilizing a Patch Pyramid Transformer to achieve effective fusion results. However, Transformers are generally less efficient than CNNs in representing local details of images. As a result,

combining CNNs and Transformers for feature extraction has emerged as a popular approach in image fusion. Representative methods include CGTF [39], DATFuse [40], SwinFusion [7], and YDTR [41], which integrate CNN before Transformer layers to benefit from the complementary strengths of both architectures. Additionally, CDDFuse [42] employs both CNNs and Transformers for separate feature extraction from input images before fusing these features to construct the final fused image.

GAN-based methods enhance the feature extraction capabilities of networks by introducing adversarial learning that contrasts the fused results with specific components of source images [2], [30], [31], [32], [33], [34], [35], [36], [37]. At present, methods in this area primarily fall into two categories: single-discriminator and dual-discriminator methods. Single-discriminator methods, while common, tend to suffer from modality imbalance, which can negatively impact the visual quality of the fusion results. In response, recent advancements such as DDcGAN [2], AttentionFGAN [31], SDDGAN [32], and GAN-FM [33] have introduced a dual-discriminator network architecture. The goal of these methods is to create efficient feature extraction networks capable of extracting high-quality features from the images to be fused, ultimately improving the quality of the fusion results. Nevertheless, these approaches often do not fully consider the effect of the fusion outcomes on subsequent visual tasks.

To address the aforementioned issues, IR-VI image fusion methods that integrate with downstream tasks have been proposed. Specifically, the most common downstream tasks include semantic segmentation [9], [10], [11] and object detection [12], [13], [14]. Initially, these methods aimed to enhance fusion quality by adding a semantic segmentation or object detection network to the fusion results. They refined the outcomes using the loss functions of these specific tasks. Although these methods show improved fusion performance, they operate under the assumption that the images being fused are precisely registered. This assumption often does not hold true in practical scenarios, implying that additional image registration algorithms are necessary for handling unregistered source images. However, achieving accurate cross-modality image registration is challenging due to the significant modality discrepancy between infrared and visible images.

B. Unregistered IR-VIS Image Fusion

To address the challenges in unregistered IR-VIS image fusion, various methods that combine registration and fusion have emerged. These approaches aim to seamlessly blend both processes to effectively fuse unregistered source images. Representative examples among these are ReCoNet [16], UMF-CMGR [17], RFNet [18], and MURF [19]. ReCoNet, for example, uses an end-to-end training approach for both registration and fusion, enhancing overall performance. However, this method does not fully account for the impact of modality discrepancies on registration, which can lead to suboptimal fusion results. UMF-CMGR, on the other hand, proposes transforming visible images into pseudo-infrared ones using an image generation network. While these transformed images are then aligned with the infrared images, the effectiveness of this method is limited by the quality of the generated images. Additionally, these methods

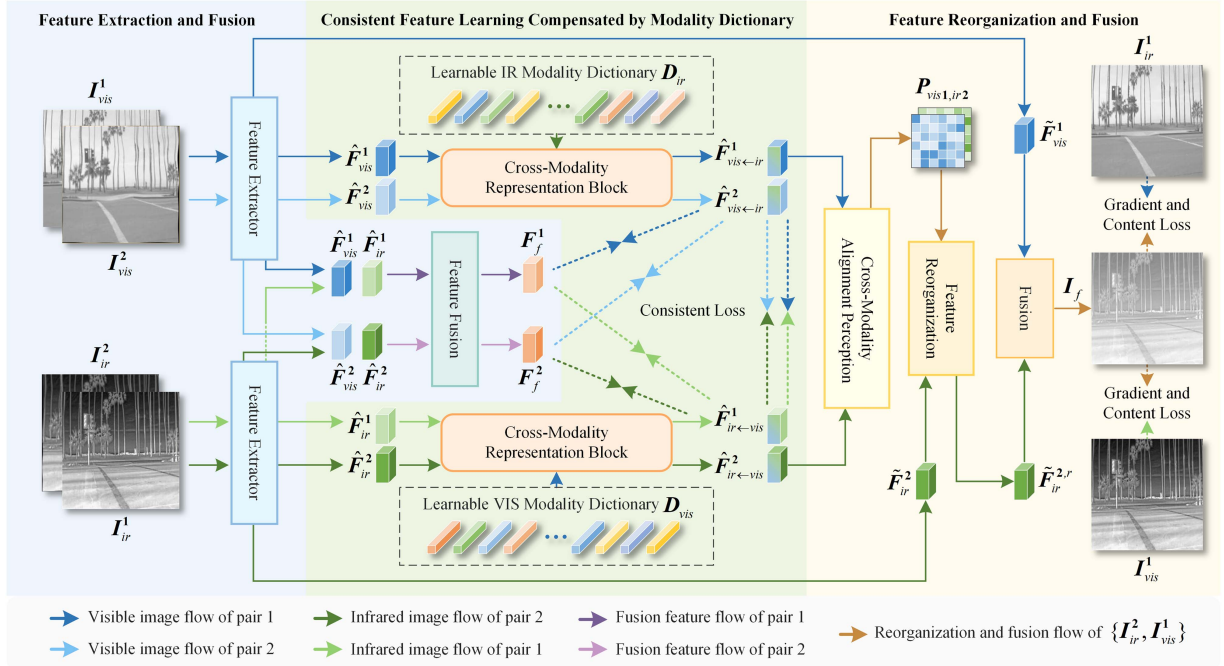


Fig. 2. Overview of the proposed method. I_{ir}^i and I_{vis}^i ($i = 1, 2$) represent two IR-VIS image pairs that have been registered, while I_{ir}^i and I_{vis}^j ($i \neq j$) form an unregistered image pair. The proposed method addresses the issue of missing specific modality information in infrared and visible images by receiving compensation information provided by a modality dictionary. This approach aims to predict the relationship matrix between pixels in infrared and visible images. When this matrix is applied to the features of the deformed image, it allows us to obtain features that are aligned with those of the fixed image, thereby facilitating their fusion. Technically, the method enhances the feature representation of unregistered images by using the fused features of registered image pairs as a guide. This approach leverages a modality dictionary to ensure that the features of an unregistered single image align with those of the fused registered images, minimizing the discrepancies across different modalities. Subsequently, an alignment perception matrix is employed to reorganize the features of the unregistered images, resulting in aligned multimodal features. These features are then fused and fed into a decoder to produce the final fused image.

are trained independently for each task, this impedes synergistic enhancements, often yielding fusion results that are less than optimal.

To enhance the fusion quality of unregistered source images, methods like RFNet and MURF view image fusion as a task subsequent to registration. They utilize the sparsity of the gradients in fusion results to enhance the performance of the registration network. However, RFNet, similar to UMF-CMGR, utilizes a generative model to transform images from one modality to another. This process aims to reduce modality discrepancies during registration. In contrast, MURF adopts a different approach by using contrastive learning to extract features common to both modalities from the input images for registration purposes. Differing from both RFNet and UMF-CMGR, SuperFusion [20] employs a shared feature encoder and consistent feature constraints. This approach facilitates the efficient extraction of modality-shared features from the source images, enhancing the subsequent registration process.

IR-VIS images are typically rich in valuable yet complementary information, but the features shared across modalities often contain limited details. Consequently, successfully establishing registration using shared features across modalities presents a significant challenge. Moreover, the above methods follow a two-stage processing paradigm, dependent on a complete and explicit registration algorithm to achieve accurate registration results. However, this two-stage approach may not be practical in environments with computational constraints. Existing methods like RFVIF [23] and IVFSR [24] attempt to address this

issue, but they fall short in handling misalignments caused by hybrid rigid-nonrigid deformation in the images to be fused. In contrast, our approach in this paper addresses both the modality discrepancy between source images with hybrid rigid-nonrigid deformation and the limitations of relying solely on shared information for registration. By directly incorporating feature alignment techniques into the image fusion process, we achieve deformation IR-VIS image fusion within a single-stage framework, thus eliminating the need for complete explicit registration.

III. THE PROPOSED METHOD

A. Overview

In Fig. 2, we illustrate the MulFS-CAP, comprising three primary components: Feature Extraction and Fusion (FEF), Consistent Feature Learning Compensated by Modality Dictionary (CFLC-MD), and Feature Reorganization and Fusion (FRF). The FEF component extracts features from pairs of registered and unregistered images, using these features from registered pairs to guide the alignment of unregistered images. Specifically, for effective supervised learning, we utilize two pairs of source images, denoted as $\{I_{ir}^i, I_{vis}^i\}_{i=1,2}$, where I_{ir}^i and I_{vis}^i represent the infrared and visible images in each pair ($i = 1, 2$). While each pair is precisely registered, the cross pairs (e.g., I_{ir}^i, I_{vis}^j , where $i \neq j$) remain unregistered due to the presence of both rigid and non-rigid deformations in one image. The registered pairs are then used to supervise the feature extraction and alignment from

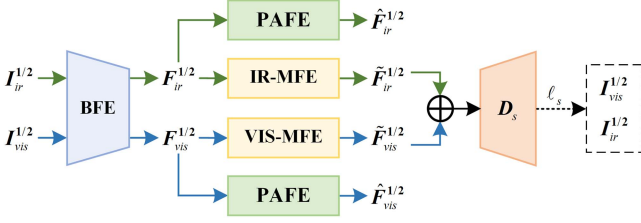


Fig. 3. Structure of feature extractor. IR-MFE and VIS-MFE do not share parameters.

the unregistered pairs. In CFLC-MD, we achieve the removal of modality discrepancies by receiving compensation information provided by the modality dictionary, effectively eliminating the disparities between infrared and visible image features. This process significantly facilitates the establishment of cross-modality correspondences between pixels. Lastly, FRF reorganizes and fuses features in their spatial positions, guided by the relationships established through Cross-Modality Alignment Perception (CMAP). Further details on these components will be provided in the subsequent sections.

B. Feature Extraction and Fusion

1) *Feature Extraction*: The feature encoder is comprised of three distinct feature extraction blocks: the Basic Feature Extraction (BFE) block, the IR/VIS Extraction (IR/VIS-MFE) block, and the Perception Alignment Feature Extraction (PAFE) block. Specifically, the BFE block includes three feature extraction layers. The IR/VIS-MFE and PAFE blocks share the same architecture, each containing seven feature extraction layers. These layers consist of a convolutional layer with a kernel size of 3×3 and a stride of 1, followed by a batch normalization layer and a ReLU activation function layer.

For the initial processing, the infrared images $I_{ir}^{1/2}$ and the visible images $I_{vis}^{1/2}$ are input into the BFE to extract shallow-level features, denoted as $F_{ir}^{1/2}$ and $F_{vis}^{1/2}$, respectively. To derive complementary deep features, these shallow-level features are then fed separately into the IR-MFE and VIS-MFE. This process yields $\tilde{F}_{ir}^{1/2}$ and $\tilde{F}_{vis}^{1/2}$, which are essential for reconstructing the final fused image. In recognition of the benefits provided by a shared-parameter encoder for achieving more consistent feature learning, we extract the infrared features, represented as $\hat{F}_{ir}^{1/2}$, and the visible features, denoted as $\hat{F}_{vis}^{1/2}$, through the shared-parameter PAFE. $\hat{F}_{ir}^{1/2}$ and $\hat{F}_{vis}^{1/2}$ are subsequently directed into two separate flows: the Feature Fusion (FF) flow and the CFLC-MD flow. The output from the FF flow serves as a supervisory signal for the CFLC-MD flow. The detailed structure of this feature extractor is depicted in Fig. 3.

It should be emphasized that maintaining the scale of the feature map in line with the original image is crucial for preserving spatial information and achieving pixel-level alignment throughout the feature extraction process. To ensure the high quality of the fusion results from $\tilde{F}_{ir}^{1/2}$ and $\tilde{F}_{vis}^{1/2}$, this study incorporates a consistency loss. This loss is specifically designed

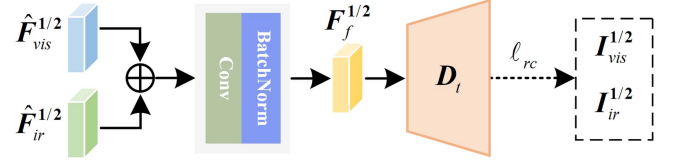


Fig. 4. Details of feature fusion in the component of feature extraction and fusion.

to constrain $\tilde{F}_{ir}^{1/2}$ and $\tilde{F}_{vis}^{1/2}$ and defined as:

$$\ell_s = \frac{1}{HW} \left(\sum_{i=1}^2 \left\| \nabla D_s(\tilde{F}_{ir}^i + \tilde{F}_{vis}^i) - \max(\nabla I_{ir}^i, \nabla I_{vis}^i) \right\|_1 + \left\| D_s(\tilde{F}_{ir}^i + \tilde{F}_{vis}^i) - I_{ir}^i \right\|_1 + \left\| D_s(\tilde{F}_{ir}^i + \tilde{F}_{vis}^i) - I_{vis}^i \right\|_1 \right), \quad (1)$$

where ∇ denotes the Sobel gradient operator, $\|\cdot\|_1$ denotes the l_1 -norm and D_s is a decoder constructed by assembling 8 feature extraction blocks. Each of these blocks comprises a convolutional layer with a kernel size of 3×3 and a stride of 1, followed by a batch normalization layer, and concluding with a ReLU activation function layer. This loss ensures that the feature extraction network extracts complementary information from the input pair of infrared and visible images, thereby contributing to the reconstruction of high-quality fused results.

2) *Feature Fusion*: In the field of unregistered image fusion, the key to achieving desired results lies in addressing the impact of pixel position offsets between the images to be fused. Traditional direct prediction methods for aligning pixel positions between infrared and visible images face challenges due to modality discrepancies, often leading to decreased precision. To address these challenges in unregistered image fusion, we utilize the fused features of $\hat{F}_{ir}^{1/2}$ and $\hat{F}_{vis}^{1/2}$ to guide the extraction of single-modality features. In the fusion process, we perform an element-wise addition of $\hat{F}_{ir}^{1/2}$ and $\hat{F}_{vis}^{1/2}$, and then fed the added result into a feature extraction block. This block, comprising a convolutional layer with a kernel size of 3×3 , a stride of 1, and a batch normalization layer, leads to the production of the fused output $F_f^{1/2}$. Such feature extraction block can enhance the fusion result by further extracting useful information from the combined features $\hat{F}_{ir}^{1/2} + \hat{F}_{vis}^{1/2}$. The details of this process are depicted in Fig. 4. To ensure the quality of $F_f^{1/2}$, we introduce an image decoder D_t and use the following loss to constrain $F_f^{1/2}$:

$$\ell_{rc} = \frac{1}{HW} \left(\sum_{i=1}^2 \left\| \nabla D_t(F_f^i) - \max(\nabla I_{ir}^i, \nabla I_{vis}^i) \right\|_1 + \left\| D_t(F_f^i) - I_{ir}^i \right\|_1 + \left\| D_t(F_f^i) - I_{vis}^i \right\|_1 \right), \quad (2)$$

where D_t is a decoder with the same architecture as D_s .

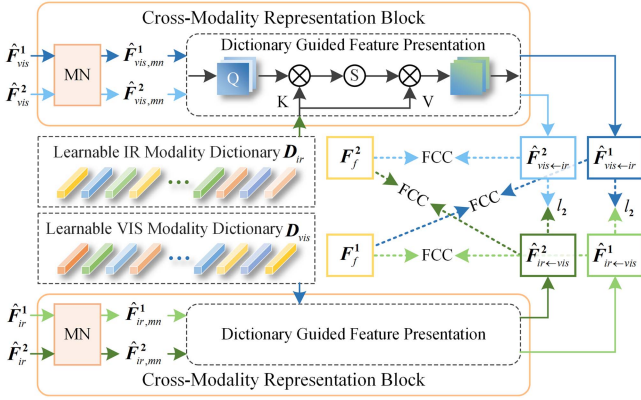


Fig. 5. Consistent feature learning compensated by modality dictionary.

C. Consistent Feature Learning

Due to the inherent modality discrepancies between infrared and visible images, directly aligning these two deformation images at the pixel level presents significant challenges. To address this issue, we suggest employing a fused feature, denoted as $F_f^{1/2}$, to supervise the feature extraction process of the input source images of one modality. However, since a single image lacks multi-modality information, achieving this alignment under the supervision of $F_f^{1/2}$ is particularly challenging. To overcome this, we develop a method named CFLC-MD, as shown in Fig. 5. This method employs a modality dictionary to compensate for the absence of specific modality information. It aims to preserve the unique features of each specific modality as much as possible. Meanwhile, it ensures that the features extracted from one modality include relevant information from the other modality. This is achieved by injecting supplementary information carried by the specific modality dictionary.

As depicted in Fig. 5, the CFLC-MD comprises two primary components: the Cross-Modality Representation Block (CMRB) and a learnable IR/VIS modality dictionary. The CMRB includes Modality Normalization (MN) and IR/VIS dictionary-guided feature presentation. The MN comprises 5 feature extraction blocks. Each block is composed of a convolutional layer with a 3×3 kernel size and a stride of 1, followed by an instance normalization layer and a ReLU activation function layer. The learnable IR/VIS modality dictionary is designed to provide complementary information, aiding in the extraction of modality-consistent features from a single image. Let D_{ir} and D_{vis} be the learnable modality dictionaries, $W_{ir/vis}^Q$, $W_{vis/ir}^K$ and $W_{vis/ir}^V$ be the linear mappings performed on $\hat{F}_{ir/vis,mn}^{1/2}$ and $D_{vis/ir}$. Then we have $Q_{ir/vis}^{1/2} = W_{ir/vis}^Q \hat{F}_{ir/vis,mn}^{1/2}$, $K_{vis/ir} = W_{vis/ir}^K D_{vis/ir}$ and $V_{vis/ir} = W_{vis/ir}^V D_{vis/ir}$. The results obtained by feeding $\hat{F}_{ir}^{1/2}$ and $\hat{F}_{vis}^{1/2}$ into MN are represented as $\hat{F}_{ir,mn}^{1/2}$ and $\hat{F}_{vis,mn}^{1/2}$. With the assistance of D_{ir} and D_{vis} , the compensated single-modality (IR/VIS) feature $\hat{F}_{ir/vis,mn}^{1/2}$ can be represented as:

$$\hat{F}_{ir \leftarrow vis / vis \leftarrow ir}^{1/2} = \text{softmax} \left(\frac{Q_{ir/vis}^{1/2} K_{vis/ir}^T}{\sqrt{d}} \right) V_{vis/ir}, \quad (3)$$

where $ir \leftarrow vis / vis \leftarrow ir$ denotes the injection of VIS/IR information into the IR/VIS modality. After compensating with the modality dictionary, the resulting features $\hat{F}_{ir \leftarrow vis}^{1/2}$ and $\hat{F}_{vis \leftarrow ir}^{1/2}$ are infused with missing modality-specific information. This step eliminates the modality discrepancies between the infrared and visible image features, thereby enhancing the informativeness of $\hat{F}_{ir \leftarrow vis}^{1/2}$ and $\hat{F}_{vis \leftarrow ir}^{1/2}$.

$F_f^{1/2}$ contains both modality-specific and modality-shared information. Therefore, using $F_f^{1/2}$ to guide the learning processes of D_{ir} and D_{vis} can ensure that they provide additional information to compensate for the limitations of single-modality features. This compensation is crucial for reducing modality discrepancies between infrared and visible image features. To achieve this goal, the features enhanced by the dictionary, which we denote as $\hat{F}_{ir \leftarrow vis / vis \leftarrow ir}^{1/2}$, should align with $F_f^{1/2}$. To ensure consistency between $\hat{F}_{ir \leftarrow vis / vis \leftarrow ir}^{1/2}$ and the fused feature $F_f^{1/2}$, which is crucial for facilitating the alignment of infrared and visible image features, we employ Feature-level Cross-Correlation (FCC) [43] to measure the correlation between $\hat{F}_{ir \leftarrow vis / vis \leftarrow ir}^{1/2}$ and $F_f^{1/2}$. As a result, the modality dictionaries D_{ir} and D_{vis} can be learned based on the following loss function:

$$\begin{aligned} \ell_{corre} = & -FCC(\hat{F}_{vis \leftarrow ir}^1, F_f^1) - FCC(\hat{F}_{ir \leftarrow vis}^1, F_f^1) \\ & - FCC(\hat{F}_{vis \leftarrow ir}^2, F_f^2) - FCC(\hat{F}_{ir \leftarrow vis}^2, F_f^2). \end{aligned} \quad (4)$$

Here, FCC is defined as:

$$FCC(X, Y) = \frac{\sum_{i,j,k} (X(i,j,k) - \bar{X}(k))(Y(i,j,k) - \bar{Y}(k))}{\sqrt{\sum_{i,j,k} (X(i,j,k) - \bar{X}(k))^2} \sqrt{\sum_{i,j,k} (Y(i,j,k) - \bar{Y}(k))^2}}, \quad (5)$$

$\bar{X}(k)$ and $\bar{Y}(k)$ denote the mean of the k th channel of X and Y , respectively.

Consistency should be maintained between $\hat{F}_{ir \leftarrow vis}^1$ and $\hat{F}_{vis \leftarrow ir}^1$, as well as between $\hat{F}_{ir \leftarrow vis}^2$ and $\hat{F}_{vis \leftarrow ir}^2$, since the missing specific-modality information has been compensated by D_{ir} and D_{vis} . Although the aforementioned FCC can provide constraints to meet this objective, there still exists a scenario where there is higher consistency between $\hat{F}_{ir \leftarrow vis / vis \leftarrow ir}^{1/2}$ and $F_f^{1/2}$, but weaker consistency between $\hat{F}_{ir \leftarrow vis}^{1/2}$ and $\hat{F}_{vis \leftarrow ir}^{1/2}$. To this end, we introduce a consistency loss based on the l_2 -norm to further optimize the modality dictionaries D_{ir} and D_{vis} :

$$\begin{aligned} \ell_{consi} = & \frac{1}{HW} \left(\left\| \hat{F}_{vis \leftarrow ir}^1 - \hat{F}_{ir \leftarrow vis}^1 \right\|_F^2 \right. \\ & \left. + \left\| \hat{F}_{vis \leftarrow ir}^2 - \hat{F}_{ir \leftarrow vis}^2 \right\|_F^2 \right). \end{aligned} \quad (6)$$

The total loss used to construct the modality dictionaries D_{ir} and D_{vis} is defined as:

$$\ell_{dic} = \ell_{corre} + \frac{1}{2} \ell_{consi}. \quad (7)$$

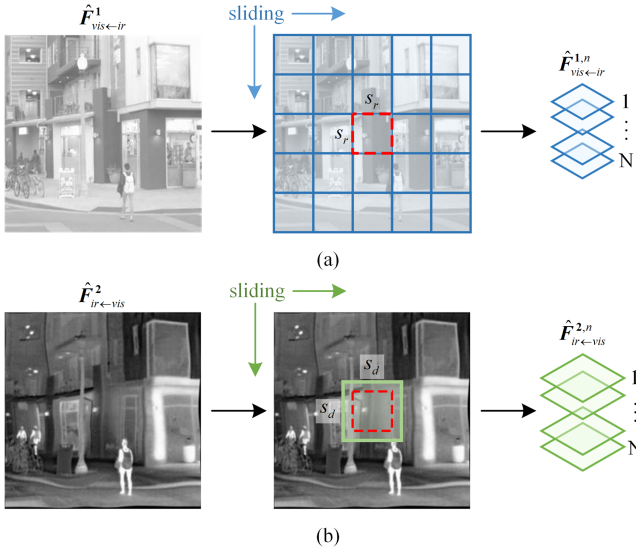


Fig. 6. Local region partition for query features and queried features. (a) The reference feature $\hat{F}_{vis \leftarrow ir}^1$ is partitioned into non-overlapping blocks of size $s_r \times s_r$. (b) A window of size $s_d \times s_d$ is utilized to partition the queried feature $\hat{F}_{ir \leftarrow vis}^2$ into blocks, while each block is centered around the region defined by the reference feature.

D. Feature Reorganization and Fusion

1) *Perceptual Region Partition*: After eliminating the modality discrepancies between $\hat{F}_{ir \leftarrow vis}^2$ and $\hat{F}_{vis \leftarrow ir}^1$, performing pixel-wise alignment of the input features using these compensated features will likely improve the alignment accuracy. However, the computational cost of directly searching for corresponding pixel features in $\hat{F}_{vis \leftarrow ir}^1$ across the entirety of $\hat{F}_{ir \leftarrow vis}^2$ is prohibitively high. In practice, spatial misalignment between infrared and visible images is typically localized, allowing the pixel search to be more efficiently conducted within a localized region.

In this paper, we designate $\hat{F}_{vis \leftarrow ir}^1$ as the reference feature, which also functions as the query feature, and $\hat{F}_{ir \leftarrow vis}^2$ as the feature with offsets, referred to as the queried feature. Our goal is to use $\hat{F}_{vis \leftarrow ir}^1$ and $\hat{F}_{ir \leftarrow vis}^2$ to learn the alignment relationship matrix. When this matrix is applied to $\hat{F}_{ir}^2$, the result aligns spatially with $\hat{F}_{vis}^1$. Concerning the partitioning of local regions, if the region size on $\hat{F}_{ir \leftarrow vis}^2$ is smaller than that on $\hat{F}_{vis \leftarrow ir}^1$, it may lead to an inability of the regions defined on $\hat{F}_{ir \leftarrow vis}^2$ to encompass all the pixels within the corresponding regions on $\hat{F}_{vis \leftarrow ir}^1$. Therefore, it is necessary for the regions on $\hat{F}_{ir \leftarrow vis}^2$ to adequately cover those on $\hat{F}_{vis \leftarrow ir}^1$.

As depicted in Fig. 6(a), we partition the reference feature $\hat{F}_{vis \leftarrow ir}^1$ into non-overlapping blocks of size $s_r \times s_r$. This results in $\hat{F}_{vis \leftarrow ir}^{1,n}$ (where $n = 1, 2, \dots, N$ and N represents the number of partition blocks). Similarly, as shown in Fig. 6(b), for the queried feature $\hat{F}_{ir \leftarrow vis}^2$, we utilize windows of size $s_d \times s_d$, which are centered around the region defined by the reference feature $\hat{F}_{vis \leftarrow ir}^1$, to identify the local region to be queried. Here, s_d is larger than s_r .

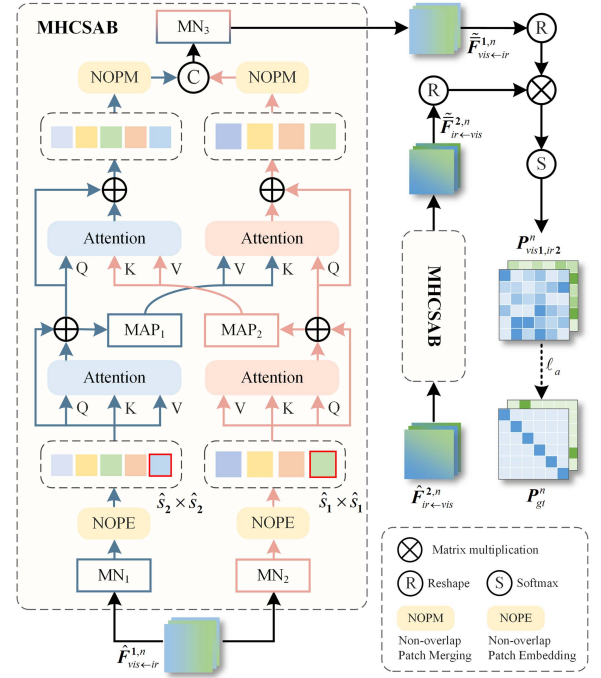


Fig. 7. Cross-modality alignment perception. MAP_l ($l = 1, 2$) is an operation that involves a sequential process of a linear mapping layer, followed by a GELU (Gaussian Error Linear Unit) activation function, and concluding with another linear mapping layer.

2) *Cross-Modality Alignment Perception*: After segmenting the feature map, which is essential for matching in localized regions, we can directly perform pixel-wise perceptual alignment within these segmented areas. However, this process faces a significant challenge: relying solely on the data provided by individual pixels in the feature map to establish pixel correspondences. To address this problem, the incorporation of multi-scale local information is verified to be highly effective. It significantly enhances the discrimination power of each pixel, aiding in more precise pixel-level feature matching. Consequently, we propose a Multi-Head Cross-Scale Aggregation Block (MHCSAB), as illustrated in Fig. 7. This block augments the feature expression of each pixel by integrating multi-scale information from its local area.

Specifically, the local region feature, denoted as $\hat{F}_{vis \leftarrow ir}^{1,n}$, is initially fed into two MN layers with the same structure and different parameters, represented as MN_l ($l = 1, 2$). Subsequently, the outputs from these layers are partitioned into local patches, each of size $\hat{s}_l \times \hat{s}_l$ (for $l = 1, 2$), where it's important to note that $\hat{s}_1 > \hat{s}_2$. These patches are then vectorized to form the feature $\mathbf{F}_{vis \leftarrow ir}^{1,n,l}$, which comprises a series of feature vectors. Following this, a linear projection is applied to $\mathbf{F}_{vis \leftarrow ir}^{1,n,l}$, yielding $\mathbf{Q}_{vis \leftarrow ir}^{1,n,l}$, $\mathbf{K}_{vis \leftarrow ir}^{1,n,l}$, and $\mathbf{V}_{vis \leftarrow ir}^{1,n,l}$. These projected components are then introduced into an attention layer, resulting in the output $\hat{\mathbf{F}}_{vis \leftarrow ir}^{1,n,l}$:

$$\hat{\mathbf{F}}_{vis \leftarrow ir}^{1,n,l} = \text{softmax} \left(\frac{\mathbf{Q}_{vis \leftarrow ir}^{1,n,l} (\mathbf{K}_{vis \leftarrow ir}^{1,n,l})^T}{\sqrt{d_l}} \right) \mathbf{V}_{vis \leftarrow ir}^{1,n,l} + \mathbf{F}_{vis \leftarrow ir}^{1,n,l} \quad (n = 1, 2, \dots, N; l = 1, 2), \quad (8)$$

where d_l denotes the dimension of $Q_{vis \leftarrow ir}^{1,n,l}$.

To enhance the interaction of information between the scales $\hat{s}_1$ and $\hat{s}_2$, we use the outputs $\tilde{Q}_{vis \leftarrow ir}^{1,n,1/2}$ from $\tilde{F}_{vis \leftarrow ir}^{1,n,1/2}$, as the Query(Q), and the outputs $\tilde{K}_{vis \leftarrow ir}^{1,n,2/1}$ and $\tilde{V}_{vis \leftarrow ir}^{1,n,2/1}$ form $\tilde{F}_{vis \leftarrow ir}^{1,n,1/2}$, processed through the mapping operation $\text{MAP}_l (l = 1, 2)$ as the Key(K) and Value(V) of the subsequent attention layer, respectively:

$$\tilde{F}_{vis \leftarrow ir}^{1,n,1} = \text{softmax} \left(\frac{\tilde{Q}_{vis \leftarrow ir}^{1,n,1} (\tilde{K}_{vis \leftarrow ir}^{1,n,2})^T}{\sqrt{d_1}} \right) \tilde{V}_{vis \leftarrow ir}^{1,n,2} + \tilde{F}_{vis \leftarrow ir}^{1,n,1}, \quad (9)$$

$$\tilde{F}_{vis \leftarrow ir}^{1,n,2} = \text{softmax} \left(\frac{\tilde{Q}_{vis \leftarrow ir}^{1,n,2} (\tilde{K}_{vis \leftarrow ir}^{1,n,1})^T}{\sqrt{d_2}} \right) \tilde{V}_{vis \leftarrow ir}^{1,n,1} + \tilde{F}_{vis \leftarrow ir}^{1,n,2}. \quad (10)$$

After the information interaction between scales $\hat{s}_1$ and $\hat{s}_2$, features $\tilde{F}_{vis \leftarrow ir}^{1,n,1}$ and $\tilde{F}_{vis \leftarrow ir}^{1,n,2}$ are infused with distinct scale-specific features, highlighting the role of cross-scale consistent features, effectively enhancing the expressive capability of $\tilde{F}_{vis \leftarrow ir}^{1,n,1}$ and $\tilde{F}_{vis \leftarrow ir}^{1,n,2}$.

To further utilize the complementary information present in $\tilde{F}_{vis \leftarrow ir}^{1,n,1}$ and $\tilde{F}_{vis \leftarrow ir}^{1,n,2}$, we execute a non-overlapping merge (NOPM) on these components. This is followed by a channel-wise concatenation. The concatenated output is then processed through the MN₃ layer, resulting in the feature $\tilde{F}_{vis \leftarrow ir}^{1,n}$, which is essential for pixel matching. Similarly, $\tilde{F}_{ir \leftarrow vis}^{2,n}$ undergoes processing via the MHCSAB block, yielding the output $\tilde{F}_{ir \leftarrow vis}^{2,n}$. Following these steps, we can derive the alignment perception matrices using both $\tilde{F}_{vis \leftarrow ir}^{1,n}$ and $\tilde{F}_{ir \leftarrow vis}^{2,n}$:

$$P_{vis1,ir2}^n = \text{Softmax}_r \left(\frac{\text{Res}(\tilde{F}_{vis \leftarrow ir}^{1,n}) \left(\text{Res}(\tilde{F}_{ir \leftarrow vis}^{2,n}) \right)^T}{\sqrt{C}} \right) \quad (n = 1, 2, \dots, N), \quad (11)$$

where “Res” denotes the “Reshape” operator, $P_{vis1,ir2}^n \in \mathbb{R}^{s_r \times s_d}$. $P_{vis1,ir2}^n(i, j)$ specifically measures the alignment perception score between the i th pixel in $\tilde{F}_{vis \leftarrow ir}^{1,n}$ and the j th pixel in $\tilde{F}_{ir \leftarrow vis}^{2,n}$. A higher score indicates a greater level of confidence in the matching between these two pixels. The notation Softmax_r refers to the application of the softmax function across the row dimensions. C signifies the number of feature channels.

To ensure the accuracy of $P_{vis1,ir2}^n$, the network parameters are optimized using relative correspondence loss ℓ_{rel} and absolute correspondence loss ℓ_{abs} [44], where ℓ_{rel} and ℓ_{abs} are defined as:

$$\ell_{rel} = \frac{1}{HWN} \sum_{n=1}^N \|P_{vis1,ir2}^n - P_{gt}^n\|_1, \quad (12)$$

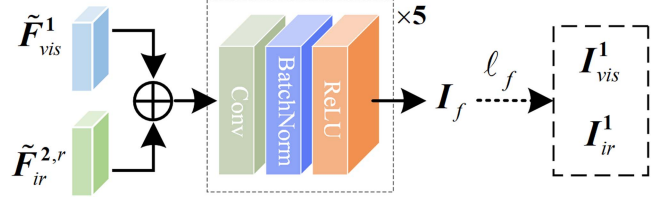


Fig. 8. Feature fusion in the component of FRF.

$$\ell_{abs} = - \sum_{n=1}^N \frac{P_{gt}^n \odot \log(P_{vis1,ir2}^n)}{\|P_{gt}^n\|_0}, \quad (13)$$

where P_{gt}^n represents the ground truth of the alignment perception matrix, which is comprised of values 0 and 1. In this matrix, if the i th pixel in $\tilde{F}_{vis \leftarrow ir}^{1,n}$ matches the j th pixel in $\tilde{F}_{ir \leftarrow vis}^{2,n}$, then $P_{gt}^n(i, j)$ is assigned the value 1; if not, it is assigned the value 0. $\|\cdot\|_0$ signifies the l_0 -norm. Additionally, $\odot$ denotes the Hadamard product. The overall loss for feature cross-modality alignment perception is defined as follows:

$$\ell_a = \ell_{rel} + \ell_{abs}. \quad (14)$$

3) *Feature Reorganization and Fusion*: The goal of feature reorganization is to spatially realign the features of infrared images with those of visible images. This is achieved by utilizing the constructed alignment perception matrices $P_{vis1,ir2}^n$. The detailed procedure is outlined as follows:

$$\tilde{F}_{ir}^{2,r,n} = \text{Res}(P_{vis1,ir2}^n \times (\text{Res}(\tilde{F}_{ir}^{2,n}))) \quad (n = 1, 2, \dots, N), \quad (15)$$

where $\text{Res}(\tilde{F}_{ir}^{2,n}) \in \mathbb{R}^{s_d \times C}$, $\tilde{F}_{ir}^{2,r,n} \in \mathbb{R}^{s_r \times s_d \times C}$. $\tilde{F}_{ir}^{2,n}$ represents the n th local region of $\tilde{F}_{ir}^{2,n}$. We proceed to reassemble $\tilde{F}_{ir}^{2,r,n}$ in order to derive $\tilde{F}_{ir}^{2,r}$. Through this reorganization, the features $\tilde{F}_{ir}^{2,r}$ are aligned with $\tilde{F}_{vis}^1$ at the feature level in their respective spatial positions. As a result, we are able to produce fused features that combine elements from both $\tilde{F}_{ir}^{2,r}$ and $\tilde{F}_{vis}^1$.

The fusion process is illustrated in Fig. 8. In this process, $\tilde{F}_{ir}^{2,r}$ and $\tilde{F}_{vis}^1$ undergo an element-wise addition before being fed into a feature extraction block. This block consists of a convolutional layer with kernel size of 3×3 and a stride of 1, followed by a batch normalization layer and a ReLU activation function layer, which work together to reconstruct the final fusion result denoted as I_f . The role of the feature extraction block is to further refine and extract valuable features, thereby enhancing the quality of the fusion results. To guarantee the effectiveness of the fusion, the process utilizes both gradient consistency loss and content consistency loss as constraints for the fusion result I_f :

$$\ell_{grad} = \frac{1}{HW} \|\nabla I_f - \max(\nabla I_{vis}^1, \nabla I_{ir}^1)\|_1, \quad (16)$$

$$\ell_{cont} = \frac{1}{HW} (\|I_f - I_{ir}^1\|_1 + \|I_f - I_{vis}^1\|_1). \quad (17)$$

Therefore, the total loss function for the fusion result I_f is defined as:

$$\ell_f = \ell_{grad} + \ell_{cont}. \quad (18)$$

Unlike the loss function in (1), this loss function primarily constrains the reconstruction process of the fused image to effectively preserve the details of the source images. The total loss function of our proposed method is defined as:

$$\ell_{total} = \ell_s + \ell_{rc} + \lambda_1 \ell_{dic} + \lambda_2 \ell_a + \lambda_3 \ell_f, \quad (19)$$

where $\lambda_i (i = 1, 2, 3)$ denotes trade-off parameters.

IV. EXPERIMENTS

A. Datasets

We trained and evaluated our proposed method using three datasets: RoadScene,¹ LLVIP [45], and MSIFT [46]. Our experimental setup followed a standard protocol in this domain. We randomly selected image pairs for training and testing as follows: From the RoadScene dataset, we used 200 IR-VIS image pairs for training and set aside 21 pairs for testing. From the LLVIP dataset, we used 201 IR-VIS image pairs for training and randomly chose 20 pairs for testing. Lastly, from the MSIFT dataset, we utilized 460 NIR-VIS (Near-Infrared-Visible) image pairs for training, with 17 pairs reserved for testing.

To enhance training effectiveness, we introduced various random deformations to image pairs within each batch. These deformations, applied equally to both images in a pair, included both rigid and non-rigid transformations. This method allows us to construct unregistered image pairs by combining the original undeformed images with their deformed counterparts. A similar strategy was employed for constructing unregistered image pairs for testing set. For model training, we applied random deformations to the registered training sample pairs in every epoch. This approach ensures that a substantial total of $S \times M$ training sample pairs were utilized throughout the model training process, where M represents the number of training epochs and S denotes the number of training samples per epoch. In our experiments, we set M to 1000.

B. Implementation Details

During the training process, each image pair is resized to 256×256 . We adopt an end-to-end training approach with a batch size of 4, and the training lasted for a total of 1000 epochs. The Adam optimizer is utilized for updating parameters throughout the training. For most modules, the initial learning rate is set at 2×10^{-4} , while for the learnable Modality Dictionary, a higher initial rate of 8×10^{-4} is employed. Notably, for the initial 160 epochs in the training of the feature cross-modality alignment perception module, a warm-up strategy is implemented, where the learning rate gradually increases from a lower value to the target learning rate of 10^{-3} . This strategy helps stabilize training by preventing large gradient updates during the early stages, when the model is still in a highly volatile state, ensuring smoother convergence and reducing the risk of divergence. The hyper-parameters λ_1 , λ_2 and λ_3 are set as $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 1$. The proposed method is implemented on an NVIDIA GeForce RTX 4090 GPU under the PyTorch framework.

¹<https://github.com/hanna-xu/RoadScene>

C. Evaluation Metrics

Since no existing metric is universally superior [47], [48], we utilized eight widely recognized image fusion evaluation metrics to objectively assess the quality of the fusion results. These include Cross entropy (Q_{CE}) [49], Peak signal-to-noise ratio (Q_{PSNR}) [50], Gradient-based fusion performance (Q_{ABF}) [51], Chen-Blum Metric (Q_{CB}) [52], Chen-Varshney metric (Q_{CV}) [53], Root mean squared error (Q_{RMSE}) [50], Structural similarity index measure (Q_{SSIM}) [54], and Sum of correlation differences (Q_{SCD}) [55]. Specifically, Q_{CE} measures the information distribution differences between the fused and source images, evaluating their similarity. Q_{PSNR} assesses image quality by calculating the ratio of maximum signal power to noise power in the fused image. $Q_{AB/F}$ evaluates how well the edge details from the source images are preserved in the fused image. Q_{CB} focuses on the incorporation of primary features from the source images into the fused image. Q_{CV} considers both human visual perception and edge saliency in the fused image. Q_{RMSE} quantifies the distortion in the fusion results by comparing the fused and source images. Q_{SSIM} examines the intensity, contrast, and structural similarity between the source and fused images. Q_{SCD} measures the correlation differences using a difference map between the fused and source images. Metrics like Q_{PSNR} , $Q_{AB/F}$, Q_{CB} , Q_{SSIM} , and Q_{SCD} are positive indicators, where higher values signify better performance. In contrast, Q_{CE} , Q_{CV} , and Q_{RMSE} are inverse indicators, with lower values indicating superior performance.

D. Comparison With State-of-The-Art Methods

In our study, we compared the proposed MulFS-CAP method with 12 leading-edge methods: CDDFuse [42], LRRNet [56], DATFuse [40], U2Fusion [3], IFCNN [5], FusionGAN [30], SemLA [22], SuperFusion [20], UMF-CMGR [17], MURF [19], IVFWSR [24] and RFVIF [23]. The comparative analysis is structured into three distinct groups. In the first group, we assess the impact of pixel misalignment on fusion results by comparing MulFS-CAP with the first six methods (CDDFuse to FusionGAN) using unregistered image pairs. For the second group, we employ CrossRAFT [57], known for its exceptional registration capabilities, to align the input images before fusion. Following this registration, the first six methods are used to fuse these aligned images. This step is designed to showcase the advantages of our approach over traditional fusion mode when working with unregistered source images. In the third group, we directly fuse unregistered image pairs using the latter six methods (SemLA to RFVIF) along with MulFS-CAP, allowing for a direct comparison of each method's ability to handle unregistered images.

1) *Performance Comparison of Directly Fusing Unregistered Images:* We conducted performance comparisons between our proposed method and several others, including CDDFuse, LRRNet, DATFuse, U2Fusion, IFCNN, and FusionGAN, using unregistered source images. The fusion results are shown in Fig. 9. As indicated by the gradient difference maps in the third column of Fig. 9, noticeable distortions and displacements are apparent in the input images. A closer examination of the magnified regions in the fusion results reveals that the comparative

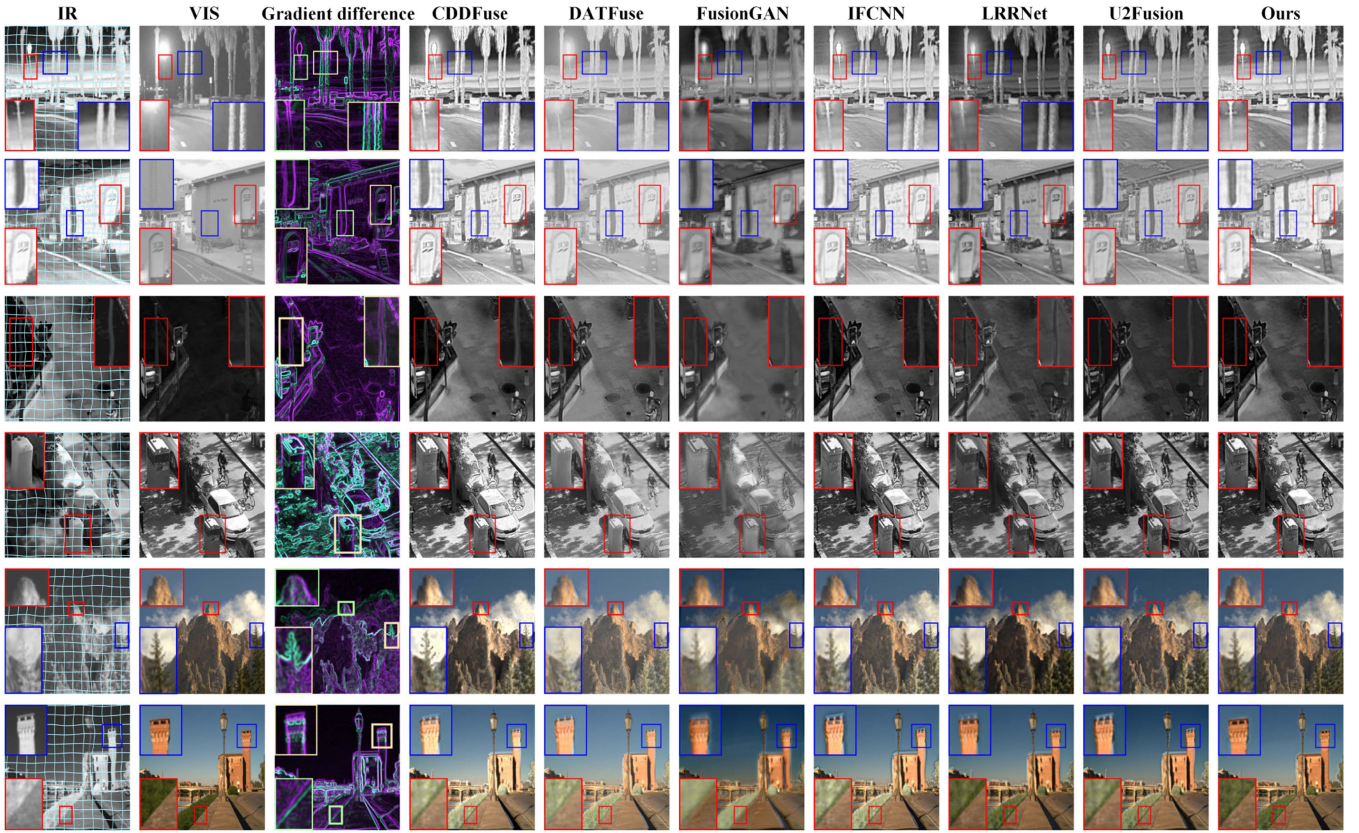


Fig. 9. Comparison of the visual effects in fusion results from various methods that directly fuse unregistered source images is presented. The first column of images shows a grid generated by the deformation field, illustrating the deformation of the images. The third column displays the gradient difference map between the infrared and visible images. For the first two columns, the first two rows of images are sourced from the RoadScene dataset, the middle two rows from the LLVIP dataset, and the final two rows from the MSIFT dataset. The fusion results from different methods are depicted in the fourth to tenth columns.

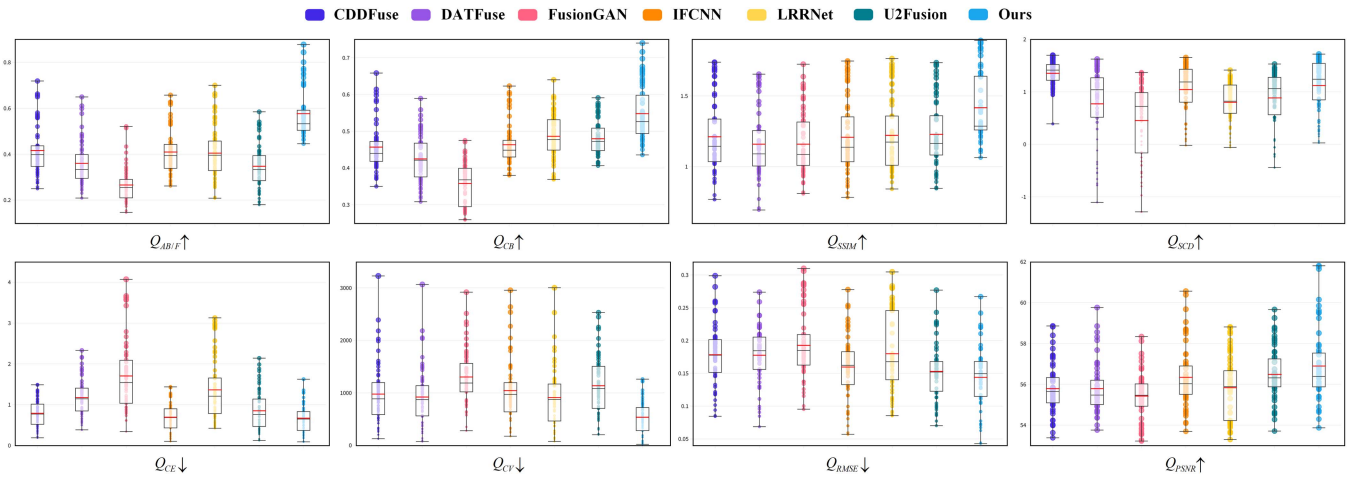


Fig. 10. Quantitative comparison of fusion methods on RoadScene, LLVIP, and MSIFT datasets using eight evaluation metrics. The x -axis represents the different methods, while the y -axis indicates their corresponding values. The upper boundary of the boxes denotes the third quartile, while the lower boundary denotes the first quartile. Within the boxes of the graphical representation, the red lines signify the mean values, and the black lines denote the median values.

methods tend to retain spatial distortions and introduce artifacts, since they are not specifically designed for unregistered images. In contrast, our proposed method effectively corrects these distortions and minimizes artifacts, resulting in more visually appealing fusion outcomes.

To further establish the superiority of our method, we evaluated the fusion results using eight commonly used evaluation metrics. For an objective demonstration of the performance of the different methods, we created box plots for all metrics, as shown in Fig. 10. These results indicate that our method

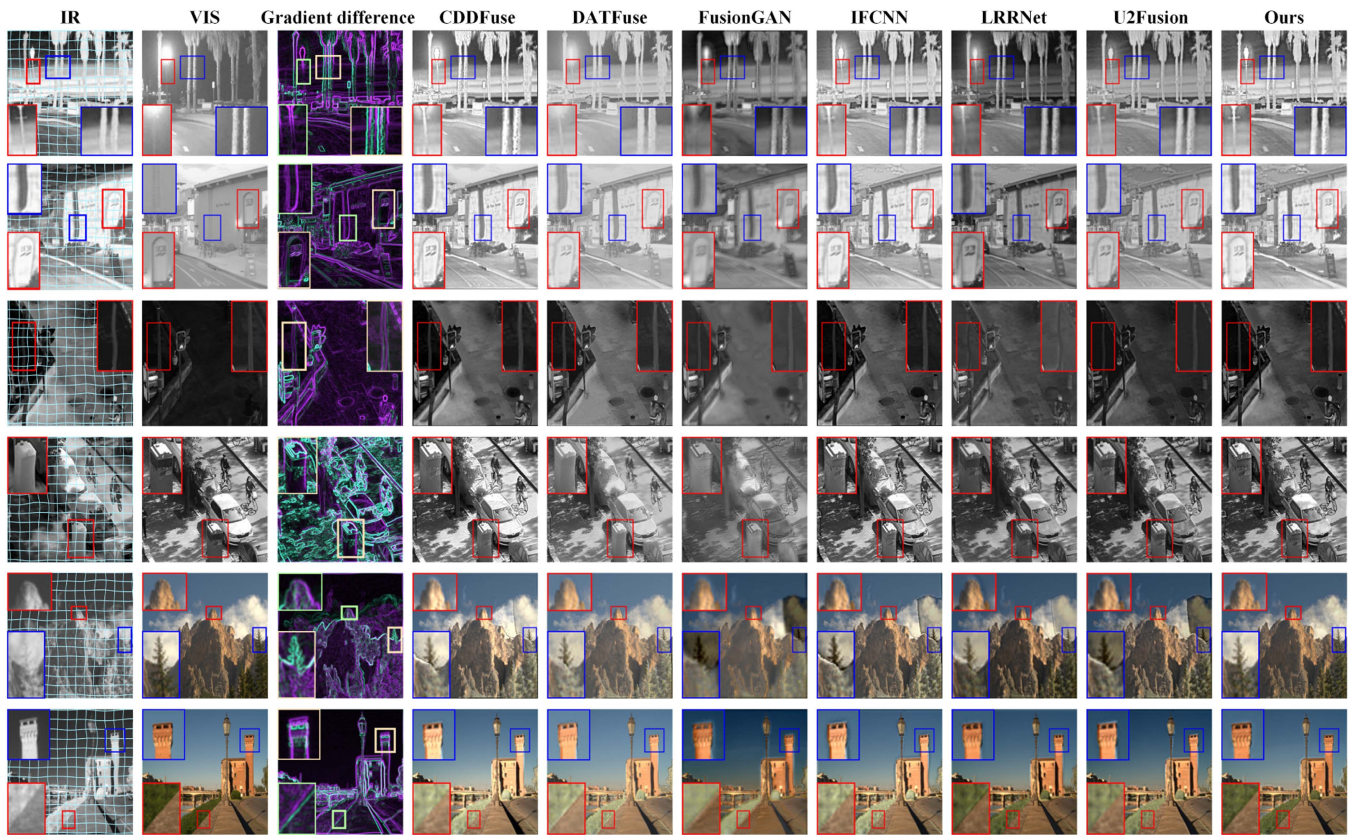


Fig. 11. Visual effect comparison of the fusion results when the registration algorithm CrossRAFT is used to register the source images first. The grid in the first column of images is generated by the deformation field, illustrating the deformation of the images. The third column of images represents the gradient difference map between the infrared and visible images. For the first two columns, the first two rows of images are sourced from the RoadScene dataset, the middle two rows from the LLVIP dataset, and the final two rows from the MSIFT dataset. The fusion results from different methods are depicted in the fourth to tenth columns.

consistently achieves the largest or second-largest mean values, demonstrating its superiority over the compared methods. Additionally, the results in Fig. 10 also highlight the stability of our method, as evidenced by its higher lower boundary values compared to other methods.

2) *Comparison With “registration+fusion”*: To address the challenge of fusing unregistered infrared and visible images, a common approach is to divide the process into two stages. Initially, an advanced registration method is employed to align the unregistered source images. This is followed by the application of a fusion algorithm to the now registered images. This approach is typically known as “Registration+Fusion”. To demonstrate the effectiveness of our proposed method compared to traditional “Registration+Fusion” methods, we utilize the high-performance CrossRAFT [57] for image registration. Afterward, we apply standard fusion methods such as CDDFuse, LRRNet, DATFuse, U2Fusion, IFCNN, and FusionGAN to perform infrared and visible image fusion. Fig. 11 depicts the visual effects of the fusion results using these different methods. These results clearly demonstrate that the introduction of the registration algorithm CrossRAFT has effectively enhanced the quality of all fusion outcomes. Moreover, a detailed comparison reveals that the method proposed in this paper excels not only in preserving the detailed information of the source images but also in achieving superior feature alignment.

Furthermore, we demonstrated the quantitative performance of each test sample pair through box plots. As illustrated in Fig. 12, the box plots for the “Registration+Fusion” methods show the distribution of evaluation scores across different test sample pairs is irregular and more dispersed. This suggests that the “Registration+Fusion” method exhibits greater performance variability with different test samples. This variability could be attributed to the unstable generalization capability of the registration method and the sensitivity of fusion methods to registration errors. In contrast, our proposed method demonstrates higher stability in performance when dealing with a diverse range of test samples. In summary, while the registration method may correct some distortions and enhance the fusion quality, the general sensitivity of these fusion algorithms to multimodal registration errors can limit further improvements in model performance.

3) *Comparison With Unregistered Image Fusion Methods*: Recent methods such as SemLA, SuperFusion, UMF-CMGR, MURF, IVFWSR, and RFVIF have been developed to tackle the challenge of fusing unregistered IR-VIS images. These methods often combine registration and fusion tasks in a two-stage paradigm, known as “Registration-Joint-Fusion” methods. Our proposed method has been both qualitatively and quantitatively compared with these methods. Fig. 13 displays the qualitative results of different methods when directly fusing unregistered

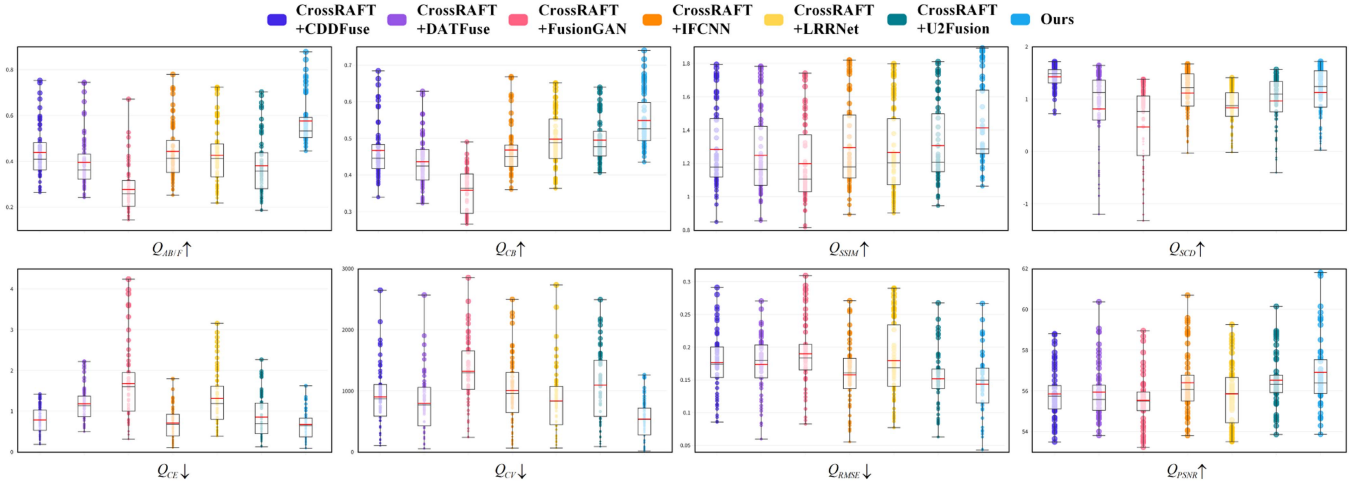


Fig. 12. Quantitative comparison of “Registration+Fusion” methods on the RoadScene, LLVIP, and MSIFT datasets across eight evaluation metrics. The x -axis represents the different methods, while the y -axis indicates their corresponding values. The upper boundary of the boxes denotes the third quartile, while the lower boundary denotes the first quartile. Within the boxes of the graphical representation, the red lines signify the mean values, and the black lines denote the median values.

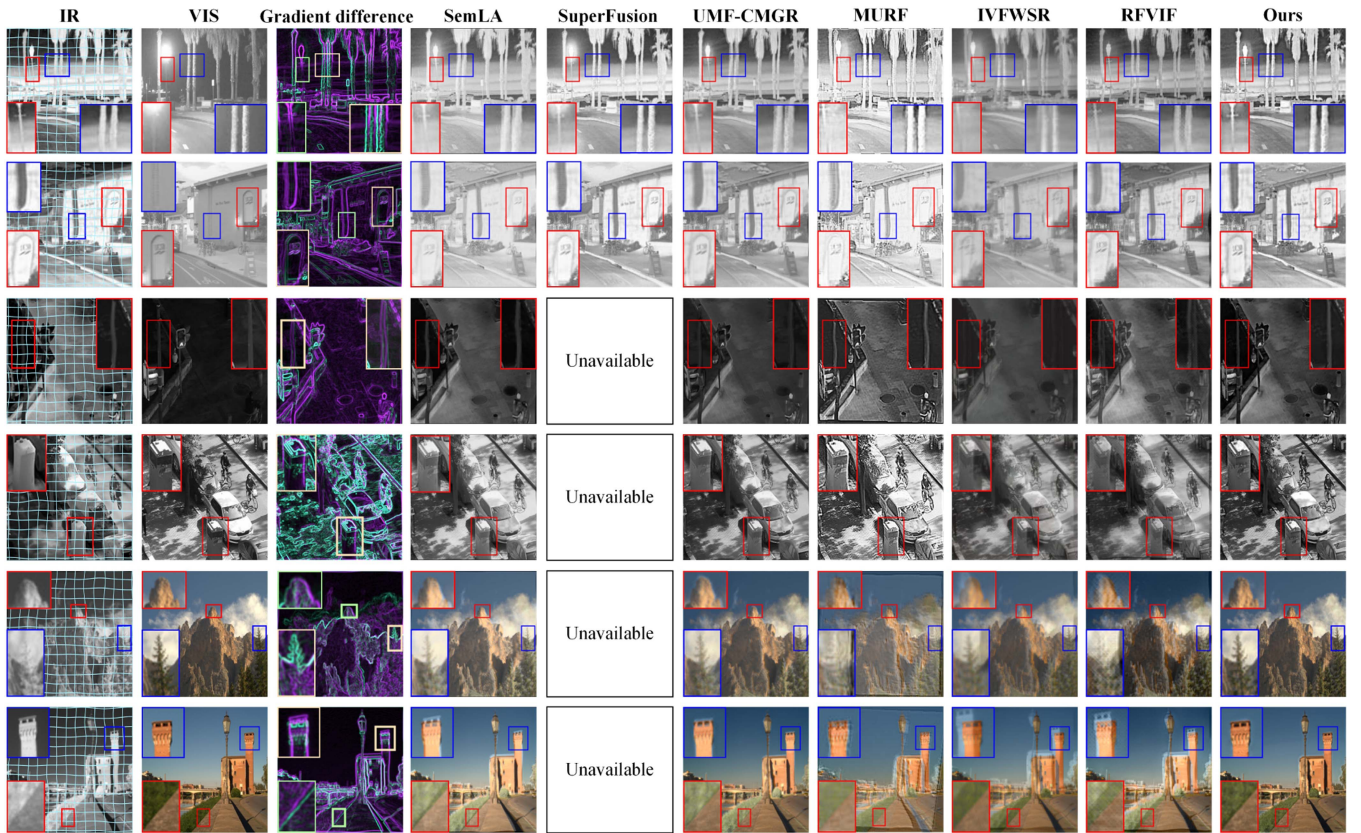


Fig. 13. Comparison of visual effects in the fusion Results of “Registration-Joint-Fusion” methods. The grid in the first column of images, generated by the deformation field, illustrates the deformation of the images. The third column displays the gradient difference map between the infrared and visible images. For the first two columns, the first two rows of images are sourced from the RoadScene dataset, the middle two rows from the LLVIP dataset, and the final two rows from the MSIFT dataset. The fusion results from different methods are depicted in the fourth to tenth columns. The fusion results of SuperFusion on the LLVIP and MSIFT datasets are unavailable; therefore, we use images labeled “unavailable” in their place.

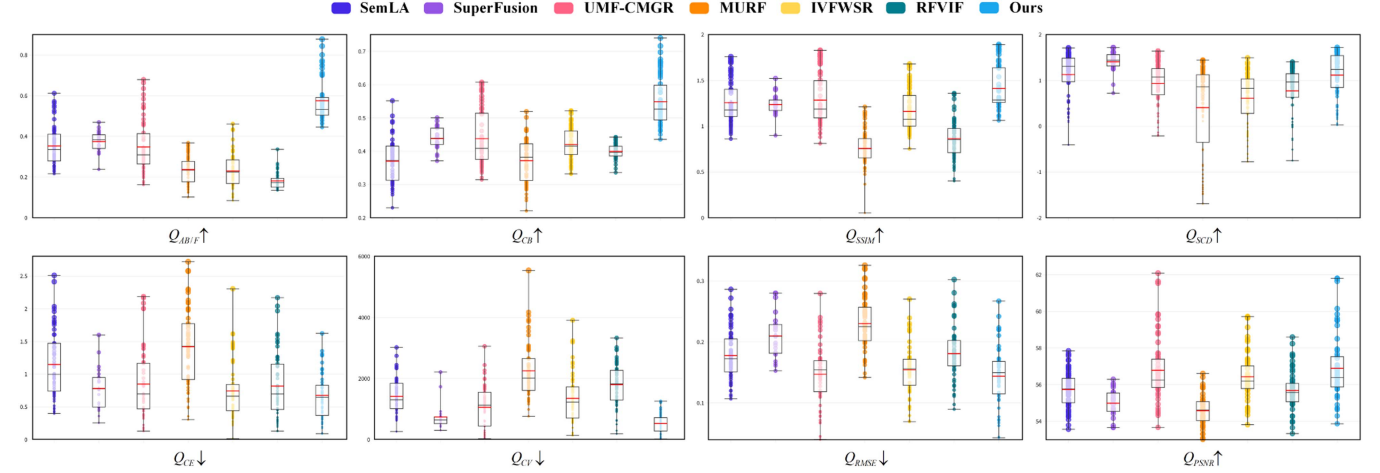


Fig. 14. Quantitative comparison with “Registration-Joint-Fusion” methods on the RoadScene, LLVIP, and MSIFT datasets on eight evaluation metrics. The x -axis represents the different methods, while the y -axis indicates their corresponding values. The upper boundary of the boxes denotes the third quartile, while the lower boundary denotes the first quartile. Within the boxes of the graphical representation, the red lines signify the mean values, and the black lines denote the median values.

source images. To visually analyze the quantitative results, box plots, as shown in Fig. 14, are used to evaluate the performance of different methods. The results in Fig. 14 reveal that our proposed method performs better across all evaluation metrics, demonstrating its superiority over the compared methods. Additionally, Fig. 14 also shows that most of the comparison methods exhibit stable performance on metrics $Q_{AB/F}$ and Q_{CB} . However, the evaluation scores for Q_{SSIM} vary significantly across these methods, while our proposed method maintains consistent performance.

4) *Further Discussion:* The superiority of our method over existing approaches largely arises from its ability to learn modality consistency features. Our proposed CFLC-MD framework not only mitigates the impact of modality diversity on cross-modality feature alignment but also preserves modality-specific information. This enhances the expressiveness of features, thereby improving registration precision. To demonstrate these advantages, we compared CFLC-MD with modules designed to address modality variance in other “Registration-Joint-Fusion” methods, specifically the SIEM module in MURF and the CP-STN in UMF-CMGR.

For a detailed analysis, we followed the method in [19] and used intensity profiles of scan lines to examine the local pixel intensity distribution in the consistency representation results. The results, as shown in Fig. 15, reveal that our method achieves highly consistent distribution along the scan lines, indicating a similar trend in pixel value variations. This suggests that CFLC-MD effectively eliminates modality discrepancies and learns discriminative features with consistent content for cross-modality registration. In contrast, the comparison methods display notable differences in pixel values between the two modalities and a lower correlation in pixel value trends, indicating persisting modality discrepancies in the learned consistency features.

To further illustrate CFLC-MD’s effectiveness, we also present histograms showing the global pixel distribution in the consistency representation results. As depicted in Fig. 15, the pixel distribution of the results generated by CFLC-MD is more concentrated, demonstrating a higher degree of overlap between

the consistency features of infrared and visible images. This further confirms that our method effectively achieves consistency representation across different modalities and addresses the issue of information inconsistency caused by modality discrepancies more efficiently than the comparison methods.

E. Ablation Study

The proposed MulFS-CAP method comprises three main components: FEF, CFLC-MD, and FRF. Within FEF, the core component is the Feature Fusion (FF), whereas in CFLC-MD, it is the learnable modality dictionary that plays a crucial role. For FRF, the MHCSAB within the CMAP is the key component. To validate the effectiveness of these components, we conducted a comprehensive series of ablation experiments. These experiments were performed on the RoadScene dataset. We performed qualitative analyses on two testing samples.

Effectiveness of FF: The FF is a key component of the FEF in our proposed MulFS-CAP. It is utilized to fuse registered infrared and visible image features, supervising the learning of modality dictionaries under the ℓ_{corre} constraint. To assess the effectiveness of FF, we conducted an experiment where FF was removed from MulFS-CAP, leaving only ℓ_{consi} for enforcing consistency constraints. For visual comparison, we input two pairs of registered source images into this simplified version of MulFS-CAP (which excludes both the CFLC-MD and feature reorganization) to generate fusion results, which are then used as pseudo ground truths.

The fusion results, as depicted in Fig. 16, reveal that the absence of FF in MulFS-CAP leads to a failure in correcting distortions and introduces issues such as artifacts and edge blurring. Additionally, as shown in Fig. 17, the visualization of the alignment perception matrices indicates that the lack of FF results in increased registration errors, attributed to the loss of discriminative information in the consistency features. This highlights the crucial role of FF in supervising the learning of cross-modality consistent features. The objective evaluation results presented in Table I further corroborate this conclusion.

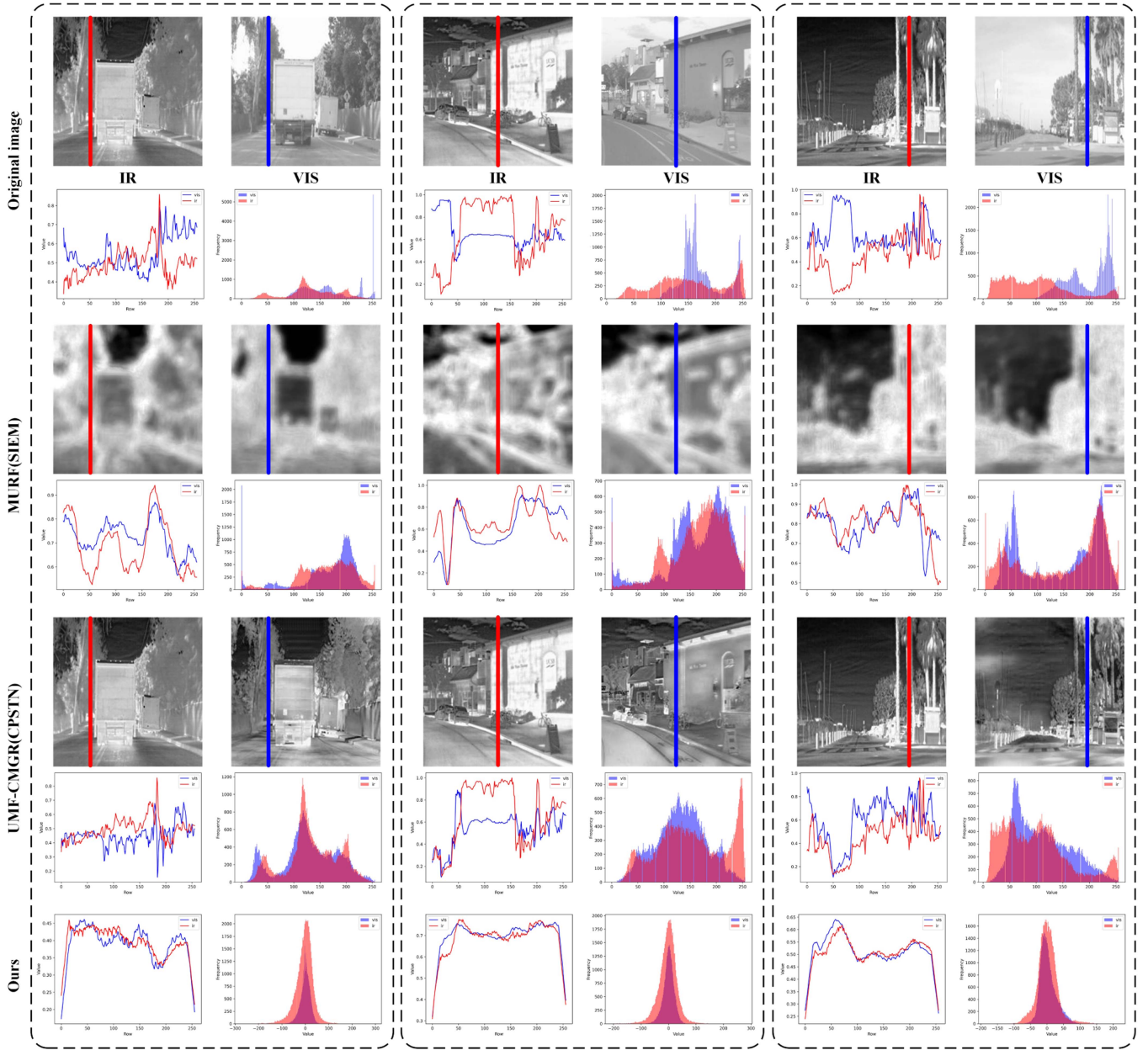


Fig. 15. Performance comparison of various methods in addressing modality discrepancies. This comparison includes dashed boxes, which represent the consistency representation results for the same pair of IR-VIS images. Within each dashed box, there are two analysis plots. On the left, the intensity profiles of the scan lines are depicted: the red line represents the intensity distribution in the infrared image, while the blue line represents that in the visible image. On the right, histograms are shown, where the red area indicates the infrared image's intensity distribution, the blue area indicates the visible image's intensity distribution, and the deep red area highlights the overlapping region between the two.

TABLE I
QUANTITATIVE RESULTS OF ABLATION STUDY

Methods	$Q_{CE} \downarrow$	$Q_{PSNR} \uparrow$	$Q_{AB/F} \uparrow$	$Q_{CB} \uparrow$	$Q_{CV} \downarrow$	$Q_{RMSE} \downarrow$	$Q_{SSIM} \uparrow$	$Q_{SCD} \uparrow$
w/o FF	0.9548	55.2995	0.2802	0.4305	1369.9378	0.1951	0.9054	1.1141
w/o LMD	0.9809	55.1705	0.3094	0.4399	979.6235	0.2009	0.9154	1.1605
w/o CFLC-MD	1.0699	55.2879	0.3028	0.4328	1306.2991	0.1955	0.9124	1.0554
w/o MHCSAB	0.7948	55.5674	0.4339	0.5305	1106.7906	0.1840	1.1962	1.1504
Ours	0.7807	55.4935	0.5271	0.5142	740.9832	0.1868	1.2576	1.4228

The results in **red** indicate the optimal.

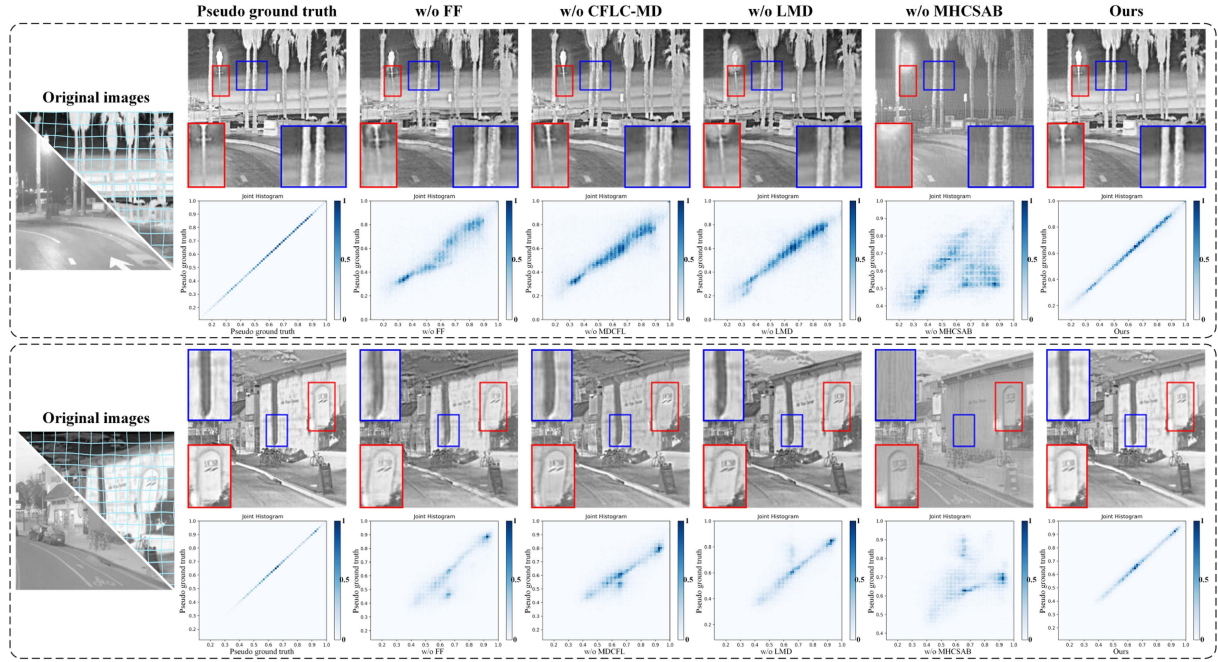


Fig. 16. Ablation studies of different components. The joint histograms in the first column display the pixel-level joint distribution between the pseudo ground truth and its corresponding intensity values. In these histograms, a completely consistent joint distribution between two images is represented as a line. The other joint histograms depict the joint probability distribution between the pseudo ground truth and the fusion results from various models. The closer these distributions are to a straight line, the fewer artifacts and deformations are present in the fusion results. This alignment indicates a closer match of the pixel intensity distribution with the pseudo ground truth, signifying better fusion quality.

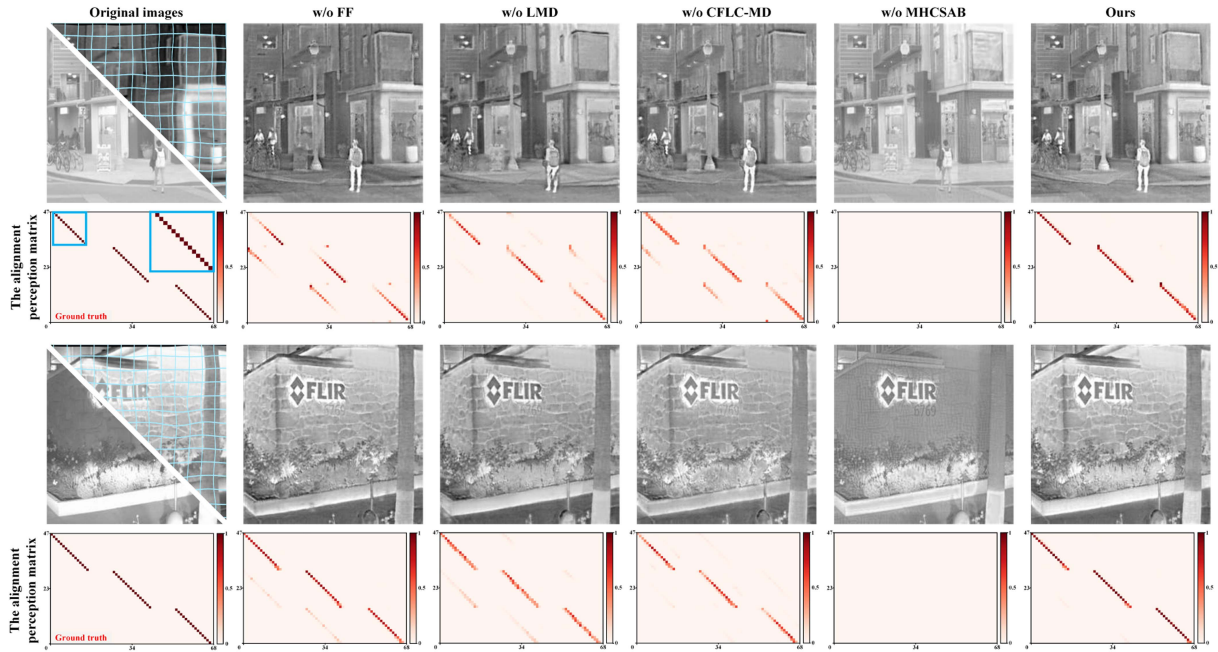


Fig. 17. Evaluation of alignment performance is conducted across different components. The X-axis in the alignment perception matrix represents the pixel coordinates in the local region of the queried image, while the Y-axis corresponds to the pixel coordinates in the local region of the reference image. If the a-th pixel in the reference image corresponds to the b-th pixel in the queried image, this correspondence is indicated by a red mark at position (a, b) in the matrix. Darker shades of color in the matrix indicate higher alignment perception scores, suggesting stronger and more reliable pixel correspondences.

Effectiveness of CFLC-MD: CFLC-MD plays a crucial role in our method, reducing the impact of modality discrepancies on cross-modality feature alignment. To evaluate its effectiveness, we conducted an ablation study by removing CFLC-MD and

directly feeding the features extracted by the feature extractor into the CMAP. As a result of removing CFLC-MD, the FF component is also omitted. The results of this ablation, including joint histograms, are displayed in Fig. 16. The results in Fig. 16

show that the model, lacking CFLC-MD, struggles to mitigate the adverse effects of modality discrepancies on feature alignment perception. This leads to spatial distortions and artifacts in the fusion results. Furthermore, the joint histograms of the fusion results from our full MulFS-CAP method are observed to be closest to those of their corresponding pseudo ground truths, indicating a high similarity in the intensity distribution of pixel values. This demonstrates that the complete MulFS-CAP model is most effective in correcting distortions and minimizing artifacts

To further assess the impact of CFLC-MD on feature alignment perception, we closely examined the local regions of the alignment perception matrices, as illustrated in Fig. 17. The findings from Fig. 17 suggest that the alignment perception matrices generated by MulFS-CAP, when CFLC-MD is absent, exhibit lower perception scores in local regions where correct correspondences are expected. Additionally, there are instances where several pixels correspond to more than one location. In such cases, these perception matrices struggle to accurately reconstruct deformed features for precise alignment. This demonstrates the critical role of CFLC-MD in MulFS-CAP. The quantitative evaluation results presented in Table I further reinforce the effectiveness of CFLC-MD.

Effectiveness of the LMD: The Learnable Modality Dictionary (LMD) is crucial in our model for providing additional information to unimodal features, ensuring their consistency with fused features across multiple modalities. To verify the effectiveness of LMD, we conducted an experiment by removing both the CMRB and LMD from the CFLC-MD. During training, the features output by the MN are subjected to consistency constraints via ℓ_{corre} and ℓ_{cons} . As evidenced by the results in Fig. 16, the model lacking LMD, similar to the one without CFLC-MD, faces difficulties in correcting distortions and suppressing artifacts. The visualization of alignment perception matrices in Fig. 17 further shows that the correspondence between pixels to be aligned becomes uncertain, leading to significant deviations from the ground truth. This highlights the vital role of the modality dictionary in mitigating modality discrepancies and aligning features between modalities.

Effectiveness of the MHCSAB: The MHCSAB used in the CMAP plays a critical role in enhancing the discrimination of individual pixels by incorporating multi-scale information, thus improving pixel-level matching accuracy. To assess the effectiveness of MHCSAB, we conducted an experiment where it was removed from CMAP, and the input features were aligned directly. According to the results in Table I, MulFS-CAP without MHCSAB shows a slight decline in fusion performance based on objective evaluation metrics. Moreover, the fusion results exhibit the introduction of unknown stripe artifacts, compromising the visual quality. The joint histogram in Fig. 16 highlights the significant discrepancies in pixel value intensity distribution between the fused results and their pseudo ground truths.

Additionally, as shown in Fig. 17, the model lacking MHCSAB fails to generate correct alignment perception matrices. This is due to the reduced feature discrimination, leading to an inability to find corresponding pixels in local neighborhoods, which also explains the scattered nature of the joint histograms in Fig. 16. However, it's noteworthy that the exclusion of

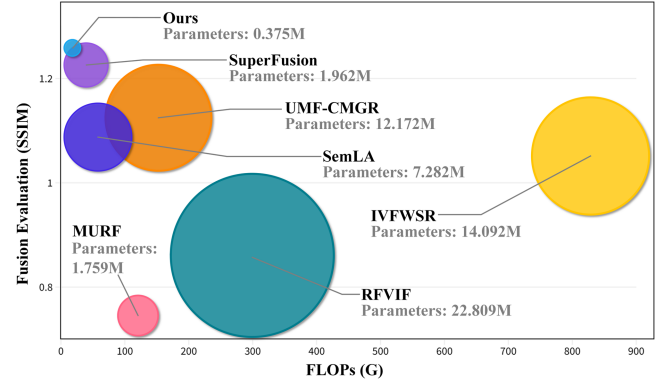


Fig. 18. Comparison of Parameter Scales and Model Performance: This graph plots the relationship between FLOPs (measured on 256×256 infrared and visible images) on the X-axis, and SSIM (Structural Similarity Index Measure) evaluation results for 21 pairs of test images from the RoadScene dataset on the Y-axis. The size of each bubble (the area of the circles) represents the number of parameters in each model.

MHCSAB does not cause a significant decrease in the objective evaluation scores of the model. This is primarily because the use of balanced alignment perception matrices for feature reorganization smooths out the details in the fusion results, as seen in the enlarged area of Fig. 16. This smoothing effect, to some extent, reduces the MSE value, subsequently enhancing the values of both PSNR and RMSE.

F. Complexity Comparison

The approach presented in this paper is pioneering, being the first to apply a one-stage method to the fusion of unregistered images. A notable advantage of this method over existing ones is its more lightweight model. To illustrate this, we compared the training parameters and FLOPs of different methods. As depicted in Fig. 18, for a pair of 256×256 images, our proposed method exhibits a clear advantage, having the least number of parameters and FLOPs while still achieving the best fusion performance. This underlines the innovation, practicality, and superiority of our proposed method's design concept.

V. CONCLUSION

This paper explores the fusion of unregistered infrared and visible images and introduces a novel one-stage method for this purpose. To address feature incompatibility in registration and fusion, we propose a consistent feature learning approach augmented with modality dictionary compensation. This method compensates for the loss of specific modality information, addresses inconsistencies in features due to modality differences between infrared and visible images, and preserves complementary information. It enables the concurrent execution of two distinct tasks in a single-stage within a unified framework. Furthermore, to align features of infrared and visible images at the level of individual pixels, we present a cross-modality alignment perception method. This method establishes relationship between different pixels in local areas, facilitating the reconstruction of registration results. Consequently, it effectively corrects distortions in detailed features and their impact on fusion results. Extensive experiments demonstrate the effectiveness and

superiority of our method over existing approaches. Our method is primarily intended for fusing infrared and visible images. However, its performance may decrease when applied to other types of image fusion, such as multi-focus image fusion, and it may also suffer from reduced performance due to domain shifts between the training and test sets. In the future, we will extend our research to include the fusion of various types of unregistered source images, such as multi-focus images, medical images, and remote sensing images, within a one-stage and unified framework with domain generalization.

REFERENCES

- [1] H. Li, Y. Cen, Y. Liu, X. Chen, and Z. Yu, "Different input resolutions and arbitrary output resolution: A meta learning-based deep framework for infrared and visible image fusion," *IEEE Trans. Image Process.*, vol. 30, pp. 4070–4083, 2021.
- [2] J. Ma, H. Xu, J. Jiang, X. Mei, and X.-P. Zhang, "DDcGAN: A dual-discriminator conditional generative adversarial network for multi-resolution image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 4980–4995, 2020.
- [3] H. Xu, J. Ma, J. Jiang, X. Guo, and H. Ling, "U2Fusion: A unified unsupervised image fusion network," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 502–518, Jan. 2022.
- [4] H. Zhang, H. Xu, Y. Xiao, X. Guo, and J. Ma, "Rethinking the image fusion: A fast unified image fusion network based on proportional maintenance of gradient and intensity," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12797–12804.
- [5] Y. Zhang, Y. Liu, P. Sun, H. Yan, X. Zhao, and L. Zhang, "IFCNN: A general image fusion framework based on convolutional neural network," *Inf. Fusion*, vol. 54, pp. 99–118, 2020.
- [6] H. Zhang and J. Ma, "SDNet: A versatile squeeze-and-decomposition network for real-time image fusion," *Int. J. Comput. Vis.*, vol. 129, no. 10, pp. 2761–2785, 2021.
- [7] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via swin transformer," *IEEE/CAA J. Automatica Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.
- [8] J. Liu, R. Dian, S. Li, and H. Liu, "SGFusion: A saliency guided deep-learning framework for pixel-level image fusion," *Inf. Fusion*, vol. 91, pp. 205–214, 2023.
- [9] L. Tang, J. Yuan, and J. Ma, "Image fusion in the loop of high-level vision tasks: A semantic-aware real-time infrared and visible image fusion network," *Inf. Fusion*, vol. 82, pp. 28–42, 2022.
- [10] L. Tang, H. Zhang, H. Xu, and J. Ma, "Rethinking the necessity of image fusion in high-level vision tasks: A practical infrared and visible image fusion network based on progressive semantic injection and scene fidelity," *Inf. Fusion*, vol. 99, 2023, Art. no. 101870.
- [11] J. Liu et al., "Multi-interactive feature learning and a full-time multi-modality benchmark for image fusion and segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 8115–8124.
- [12] W. Zhao, S. Xie, F. Zhao, Y. He, and H. Lu, "MetaFusion: Infrared and visible image fusion via meta-feature embedding from object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 13955–13965.
- [13] J. Liu et al., "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5802–5811.
- [14] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "DetFusion: A detection-driven infrared and visible image fusion network," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 4003–4011.
- [15] R. Liu, Z. Liu, J. Liu, X. Fan, and Z. Luo, "A task-guided, implicitly-searched and meta-initialized deep model for image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6594–6609, Oct. 2024.
- [16] Z. Huang, J. Liu, X. Fan, R. Liu, W. Zhong, and Z. Luo, "Reconet: Recurrent correction network for fast and efficient multi-modality image fusion," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 539–555.
- [17] D. Wang, J. Liu, X. Fan, and R. Liu, "Unsupervised misaligned infrared and visible image fusion via cross-modality image generation and registration," in *Proc. Int. Joint Conf. Artif. Intell.*, 2022, pp. 3508–3515.
- [18] H. Xu, J. Ma, J. Yuan, Z. Le, and W. Liu, "RFNet: Unsupervised network for mutually reinforcing multi-modal image registration and fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 19679–19688.
- [19] H. Xu, J. Yuan, and J. Ma, "MURF: Mutually reinforcing multi-modal image registration and fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12148–12166, Oct. 2023.
- [20] L. Tang, Y. Deng, Y. Ma, J. Huang, and J. Ma, "SuperFusion: A versatile image registration and fusion network with semantic awareness," *IEEE/CAA J. Automatica Sinica*, vol. 9, no. 12, pp. 2121–2137, Dec. 2022.
- [21] D. Wang, J. Liu, L. Ma, R. Liu, and X. Fan, "Improving misaligned multi-modality image fusion with one-stage progressive dense registration," 2023, *arXiv:2308.11165*.
- [22] H. Xie et al., "Semantics lead all: Towards unified image registration and fusion from a semantic perspective," *Inf. Fusion*, vol. 98, Art. no. 101835, 2023.
- [23] H. Li, J. Zhao, J. Li, Z. Yu, and G. Lu, "Feature dynamic alignment and refinement for infrared-visible image fusion: Translation robust fusion," *Inf. Fusion*, vol. 95, pp. 26–41, 2023.
- [24] H. Li, J. Liu, Y. Zhang, and Y. Liu, "A deep learning framework for infrared and visible image fusion without strict registration," *Int. J. Comput. Vis.*, vol. 132, pp. 1625–1644, 2024.
- [25] H. Li and X.-J. Wu, "DenseFuse: A fusion approach to infrared and visible images," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2614–2623, May 2019.
- [26] H. Li, X.-J. Wu, and T. Durrani, "NestFuse: An infrared and visible image fusion architecture based on nest connection and spatial/channel attention models," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9645–9656, Dec. 2020.
- [27] H. Li, X.-J. Wu, and J. Kittler, "RFN-NEST: An end-to-end residual fusion network for infrared and visible images," *Inf. Fusion*, vol. 73, pp. 72–86, 2021.
- [28] J. Liu, X. Fan, J. Jiang, R. Liu, and Z. Luo, "Learning a deep multi-scale feature ensemble and an edge-attention guidance for image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 1, pp. 105–119, Jan. 2022.
- [29] L. Jian, X. Yang, Z. Liu, G. Jeon, M. Gao, and D. Chisholm, "SEDRFuse: A symmetric encoder-decoder with residual block network for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 5002215.
- [30] J. Ma, W. Yu, P. Liang, C. Li, and J. Jiang, "FusionGAN: A generative adversarial network for infrared and visible image fusion," *Inf. Fusion*, vol. 48, pp. 11–26, 2019.
- [31] J. Li, H. Huo, C. Li, R. Wang, and Q. Feng, "AttentionFGAN: Infrared and visible image fusion using attention-based generative adversarial networks," *IEEE Trans. Multimedia*, vol. 23, pp. 1383–1396, 2021.
- [32] H. Zhou, W. Wu, Y. Zhang, and J. Ma, "Semantic-supervised infrared and visible image fusion via a dual-discriminator generative adversarial network," *IEEE Trans. Multimedia*, vol. 25, pp. 635–648, 2023.
- [33] H. Zhang, J. Yuan, X. Tian, and J. Ma, "GAN-FM: Infrared and visible image fusion using GAN with full-scale skip connection and dual markovian discriminators," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 1134–1147, 2021.
- [34] J. Ma, H. Zhang, Z. Shao, P. Liang, and H. Xu, "GANMcC: A generative adversarial network with multiclassification constraints for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 5005014.
- [35] Y. Yang, J. Liu, S. Huang, W. Wan, W. Wen, and J. Guan, "Infrared and visible image fusion via texture conditional generative adversarial network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 12, pp. 4771–4783, Dec. 2021.
- [36] J. Ma et al., "Infrared and visible image fusion via detail preserving adversarial learning," *Inf. Fusion*, vol. 54, pp. 85–98, 2020.
- [37] Y. Fu, X.-J. Wu, and T. Durrani, "Image fusion based on generative adversarial network consistent with perception," *Inf. Fusion*, vol. 72, pp. 110–125, 2021.
- [38] J. Liu, R. Lin, G. Wu, R. Liu, Z. Luo, and X. Fan, "CoCoNet: Coupled contrastive learning network with multi-level feature ensemble for multi-modality image fusion," *Int. J. Comput. Vis.*, vol. 132, no. 5, pp. 1748–1775, 2024.
- [39] J. Li, J. Zhu, C. Li, X. Chen, and B. Yang, "CGTF: Convolution-guided transformer for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, Art. no. 5012314.
- [40] W. Tang, F. He, Y. Liu, Y. Duan, and T. Si, "DATFuse: Infrared and visible image fusion via dual attention transformer," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 7, pp. 3159–3172, Jul. 2023.

- [41] W. Tang, F. He, and Y. Liu, "YDTR: Infrared and visible image fusion via Y-shape dynamic transformer," *IEEE Trans. Multimedia*, vol. 25, pp. 5413–5428, 2023.
- [42] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 5906–5916.
- [43] J. Sarvaiya, S. Patnaik, and S. Bombaywala, "Image registration by template matching using normalized cross-correlation," in *Proc. 2009 Int. Conf. Adv. Comput., Control, Telecommun. Technol.*, 2009, pp. 819–822.
- [44] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 8922–8931.
- [45] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops*, 2021, pp. 3496–3504.
- [46] M. Brown and S. Süsstrunk, "Multi-spectral sift for scene category recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2011, pp. 177–184.
- [47] X. Zhang, P. Ye, and G. Xiao, "VIFB: A visible and infrared image fusion benchmark," in *Proc. 2020 IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2020, pp. 468–478.
- [48] Y. Liu, Z. Qi, J. Cheng, and X. Chen, "Rethinking the effectiveness of objective evaluation metrics in multi-focus image fusion: A statistic-based approach," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5806–5819, Aug. 2024.
- [49] D. Bulanon, T. Burks, and V. Alchanatis, "Image fusion of visible and thermal images for fruit detection," *Biosyst. Eng.*, vol. 103, no. 1, pp. 12–22, 2009.
- [50] P. Jagalingam and A. V. Hegde, "A review of quality metrics for fused image," *Aquatic Procedia*, vol. 4, pp. 133–142, 2015.
- [51] C. S. Xydeas et al., "Objective image fusion performance measure," *Electron. Lett.*, vol. 36, no. 4, pp. 308–309, 2000.
- [52] Y. Chen and R. S. Blum, "A new automated quality assessment algorithm for image fusion," *Image Vis. Comput.*, vol. 27, no. 10, pp. 1421–1432, 2009.
- [53] H. Chen and P. K. Varshney, "A human perception inspired quality metric for image fusion based on regional information," *Inf. Fusion*, vol. 8, no. 2, pp. 193–207, 2007.
- [54] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [55] V. Aslantas and E. Bendes, "A new image quality metric for image fusion: The sum of the correlations of differences," *AEU - Int. J. Electron. Commun.*, vol. 69, no. 12, pp. 1890–1896, 2015.
- [56] H. Li, T. Xu, X.-J. Wu, J. Lu, and J. Kittler, "LRRNet: A novel representation learning guided fusion network for infrared and visible images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 11040–11052, Sep. 2023.
- [57] S. Zhou, W. Tan, and B. Yan, "Promoting single-modal optical flow network for diverse cross-modal flow estimation," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 3562–3570.



Huafeng Li (Member, IEEE) received the MS degrees in applied mathematics major from Chongqing University, in 2009 and the PhD degree in control theory and control engineering major from Chongqing University, in 2012. He is currently a professor with the School of Information Engineering and Automation, Kunming University of Science and Technology, China. His research interests include image processing, computer vision, and information fusion. He has authored or coauthored more than 50 scientific articles in CVPR, IJCV, AAAI, ACMMM, ICME,

IEEE Transactions on Image Processing, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Circuits and Systems for Video Technology*, *IEEE Transactions on Geoscience and Remote Sensing*, *IEEE Transactions on Computational Imaging*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Emerging Topics in Computational Intelligence*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Instrumentation and Measurement*, *IEEE/CAA Journal of Automatica Sinica*, PR, INFFUS, NeuNet, INS, ESWA, KBS, etc.



Zengyi Yang received the BE degree from the Kunming University of Science and Technology, Kunming, China, in 2022. He is currently working toward the MEng degree with Kunming University of Science and Technology. His research interests include computer vision, image fusion, and deep learning.



Yafei Zhang received the PhD degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2008. She is currently an associate professor with the College of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, China. Her main research interests include image processing and pattern recognition.



Wei Jia (Member, IEEE) received the BSc degree in informatics from Central China Normal University, Wuhan, China, in 1998, the MSc degree in computer science from the Hefei University of Technology, Hefei, China, in 2004, and the PhD degree in pattern recognition and intelligence systems from the University of Science and Technology of China, Hefei, in 2008. He was a research assistant and an associate professor with the Hefei Institutes of Physical Science, Chinese Academy of Science, from 2008 to 2016. He is currently a full professor with the

School of Computer Science and Information Engineering, Hefei University of Technology. His research interests include computer vision, biometrics, pattern recognition, image processing, and machine learning.



Zhengtao Yu received the PhD degree in computer application technology from the Beijing Institute of Technology, Beijing, China, in 2005. He is currently a professor with the School of Information Engineering and Automation, Kunming University of Science and Technology, China. His main research interests include natural language process, image processing, and machine learning.



Yu Liu (Senior Member, IEEE) received the BS and PhD degrees from the Department of Automation, University of Science and Technology of China, in 2011 and 2016, respectively. He is currently a full professor in the Department of Biomedical Engineering with the Hefei University of Technology. His research interests include image processing, computer vision, information fusion and machine learning. In particular, He is interested in image fusion, image restoration, visual recognition and deep learning. He is serving as an Editorial Board Member for *Information Fusion*, and an associate editor for *IEEE Signal Processing Letters*. He was a recipient of the IEEE Instrumentation and Measurement Society Andi Chi Best Paper Award, in 2020 and the IET Image Processing Premium (Best Paper) Award, in 2017. He was identified as a Clarivate Highly Cited Researcher, in 2023 and 2024.

He was a recipient of the IEEE Instrumentation and Measurement Society Andi Chi Best Paper Award, in 2020 and the IET Image Processing Premium (Best Paper) Award, in 2017. He was identified as a Clarivate Highly Cited Researcher, in 2023 and 2024.